T19 인터포저 위 광학 집적과 링크 효율

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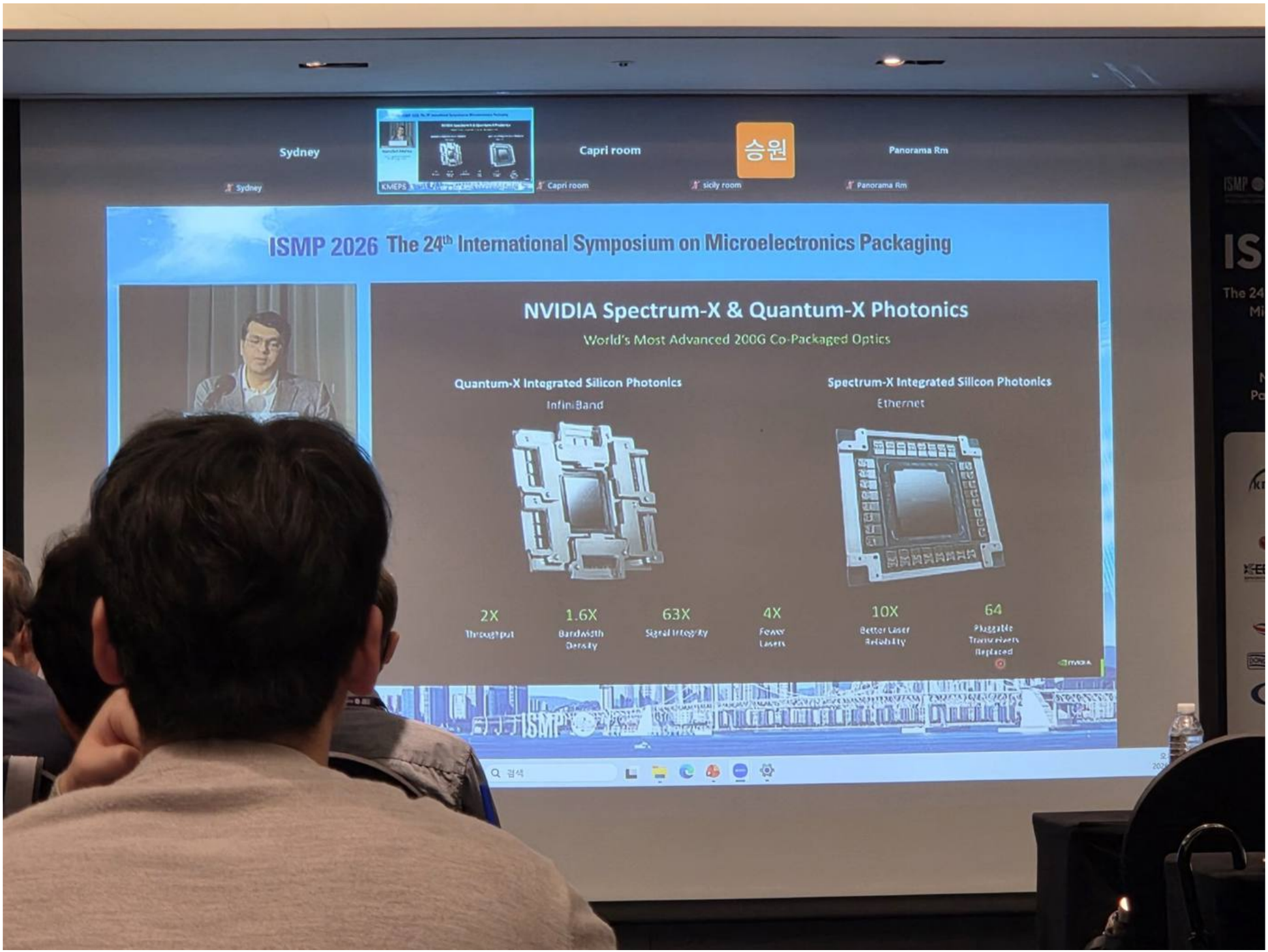

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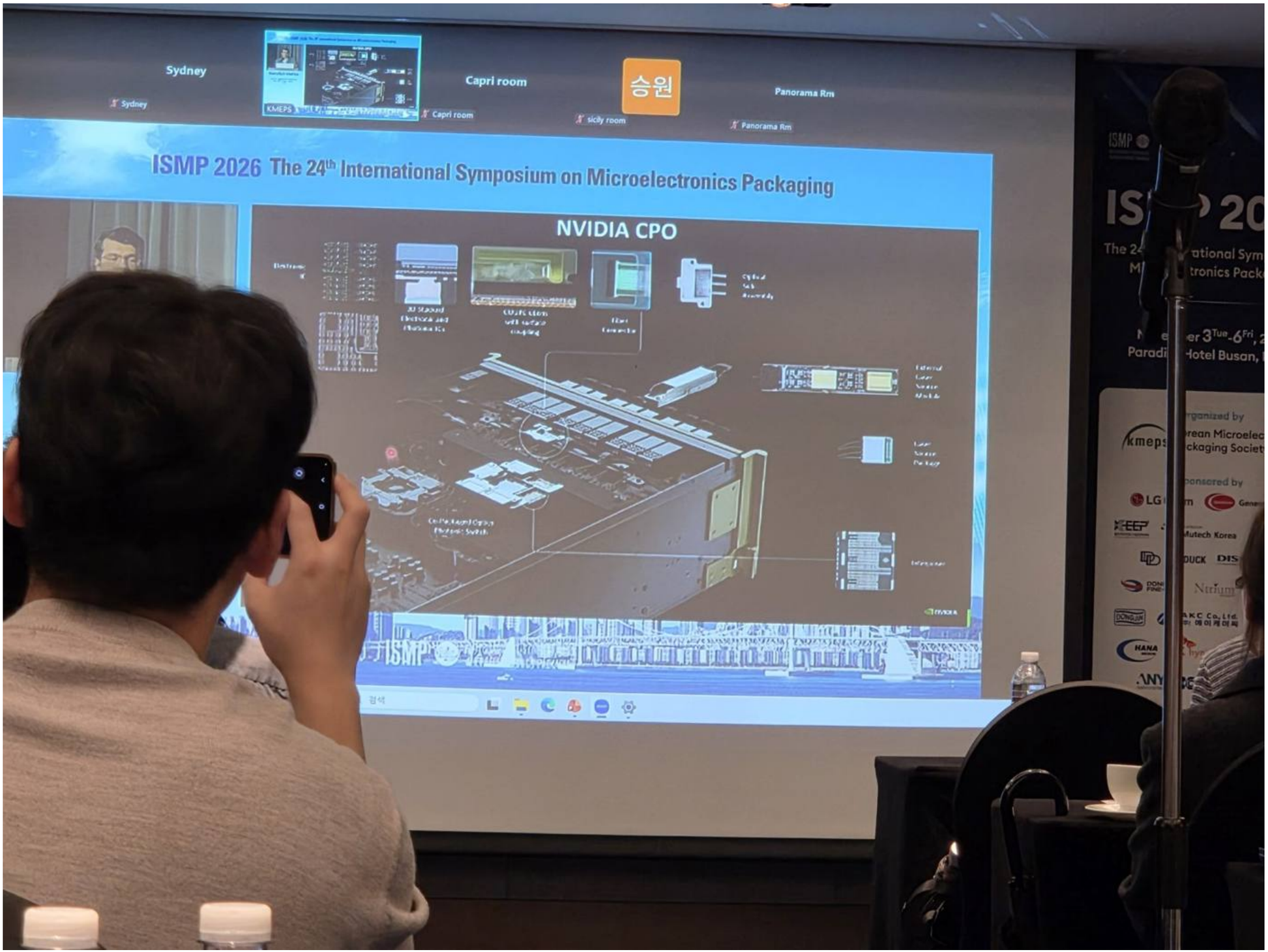

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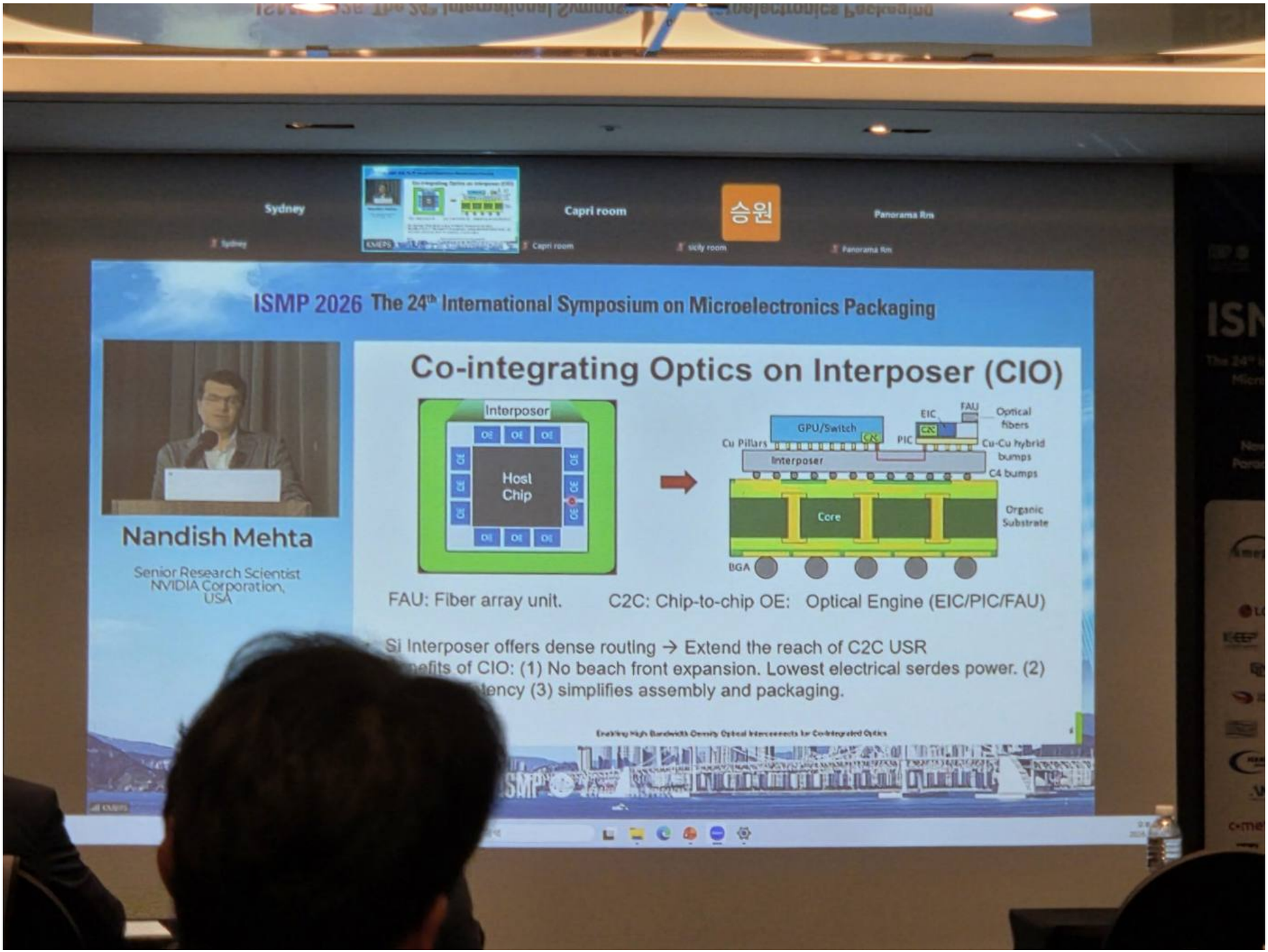

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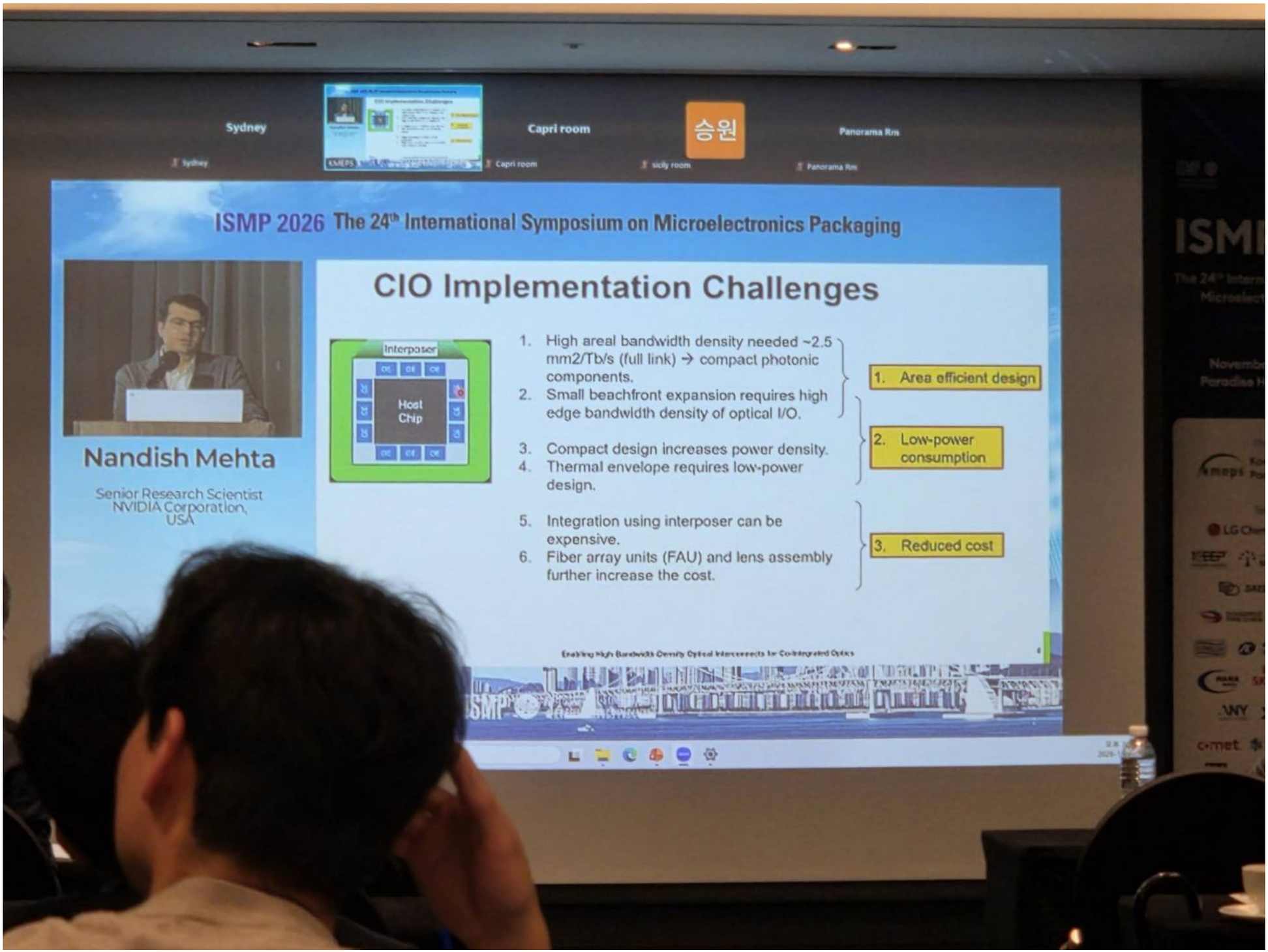

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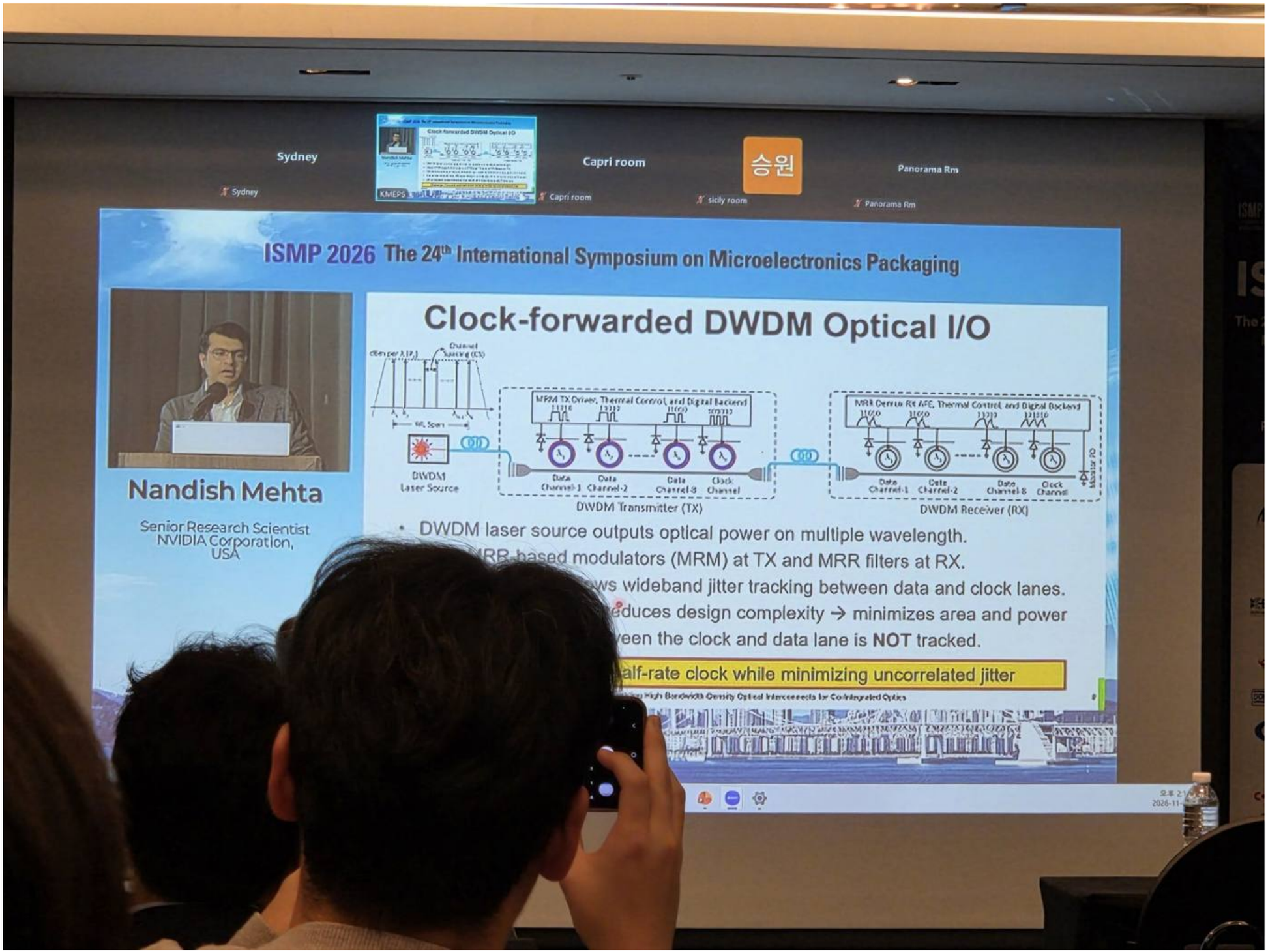

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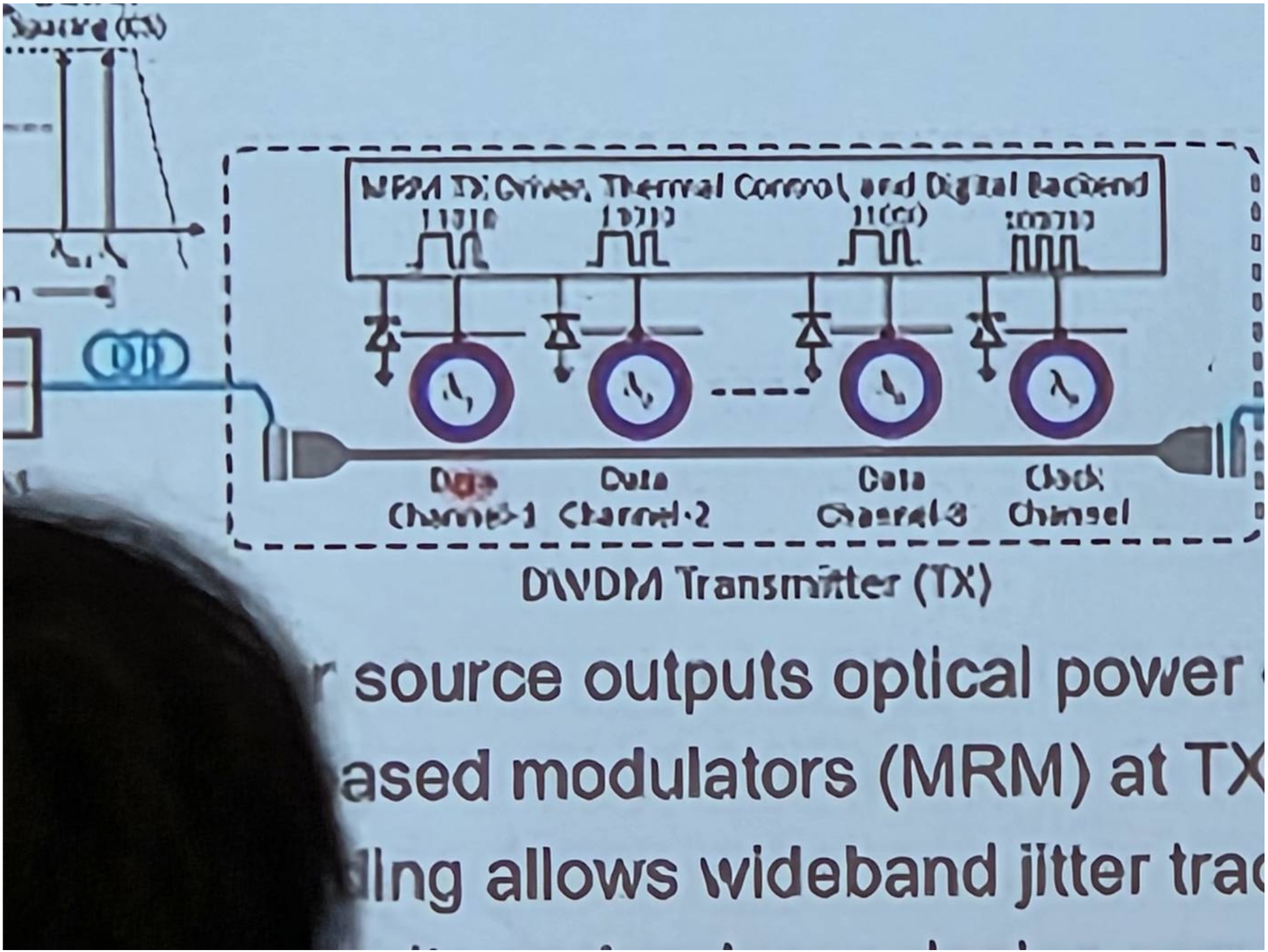

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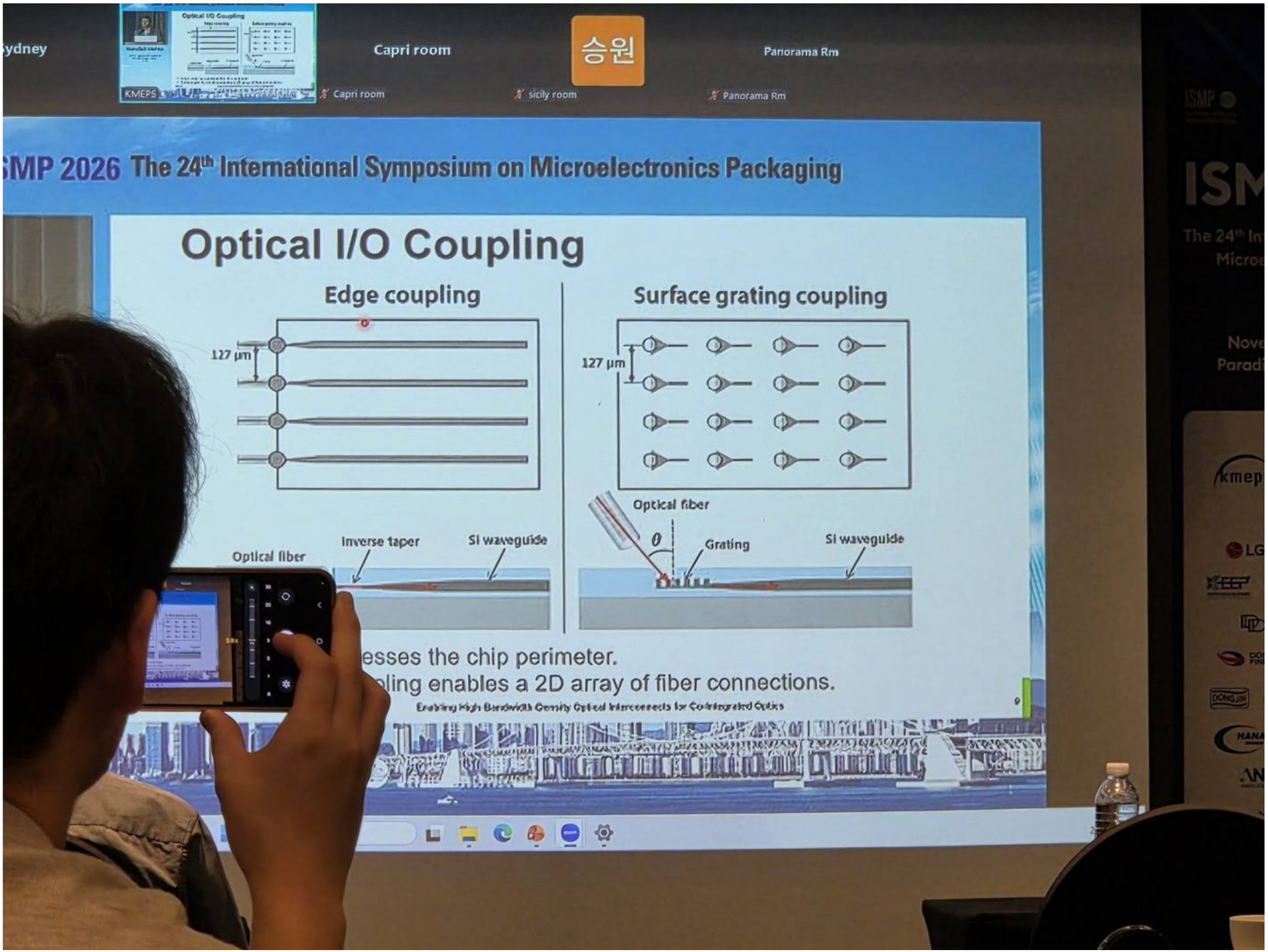

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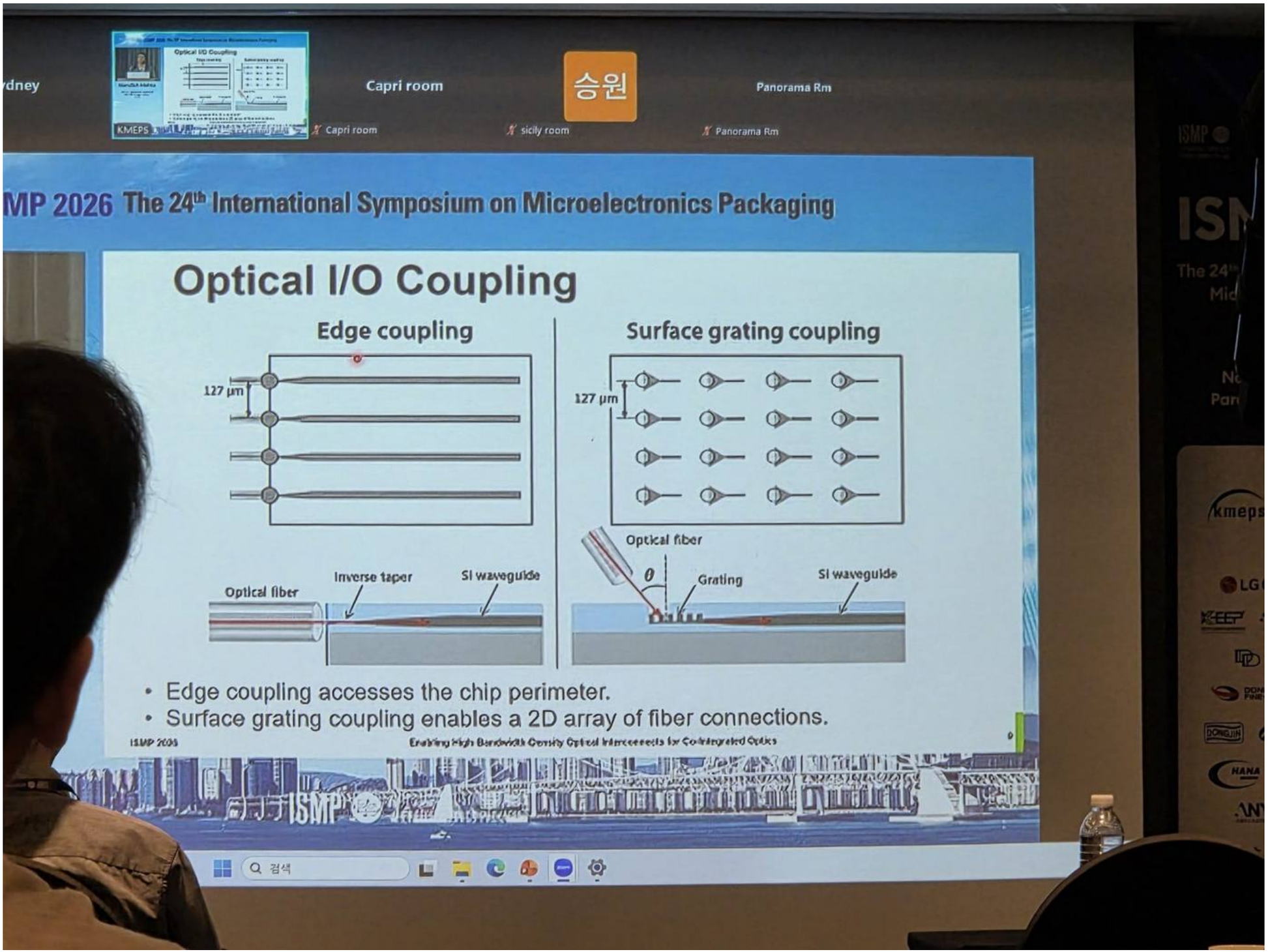

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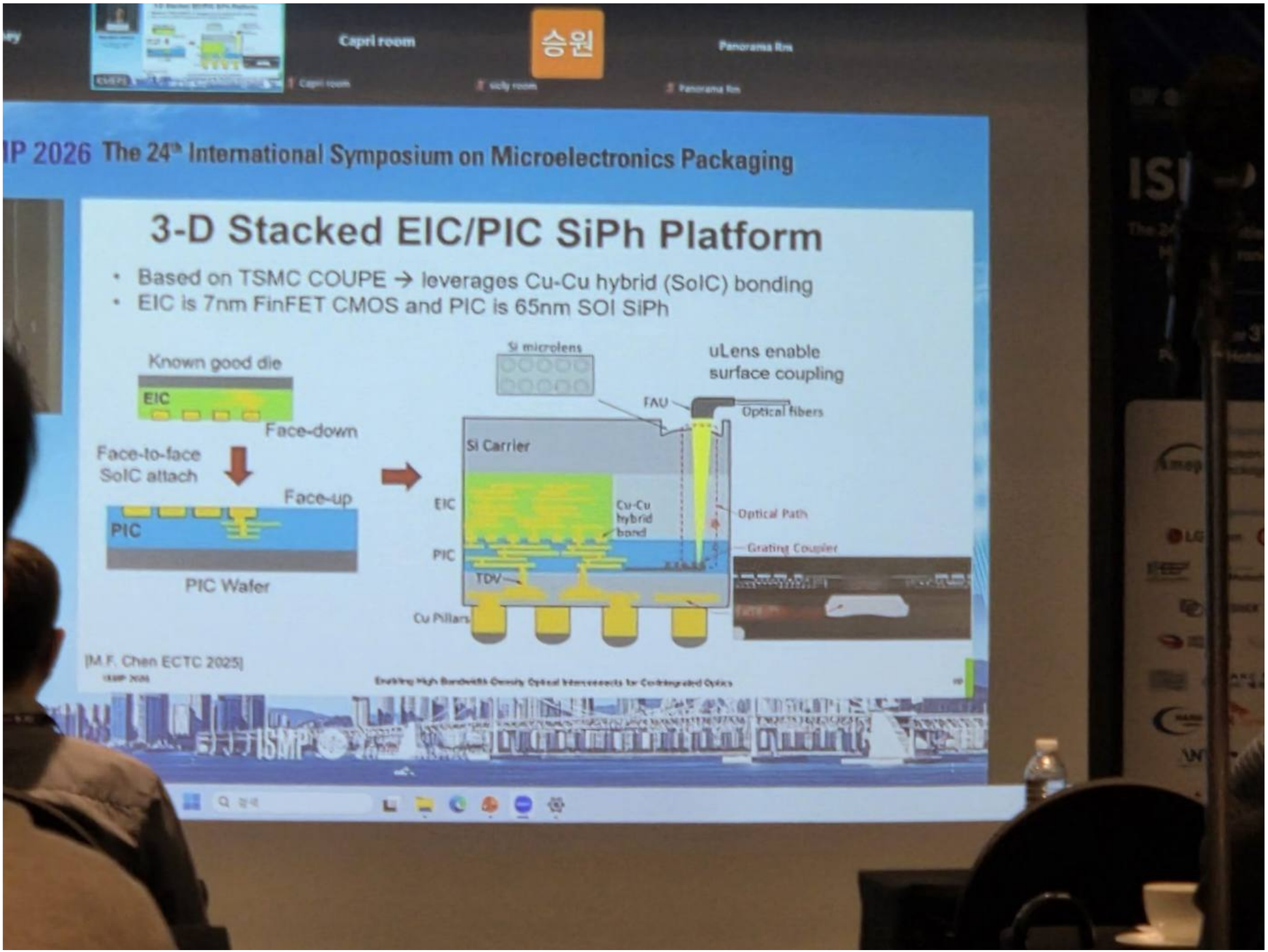

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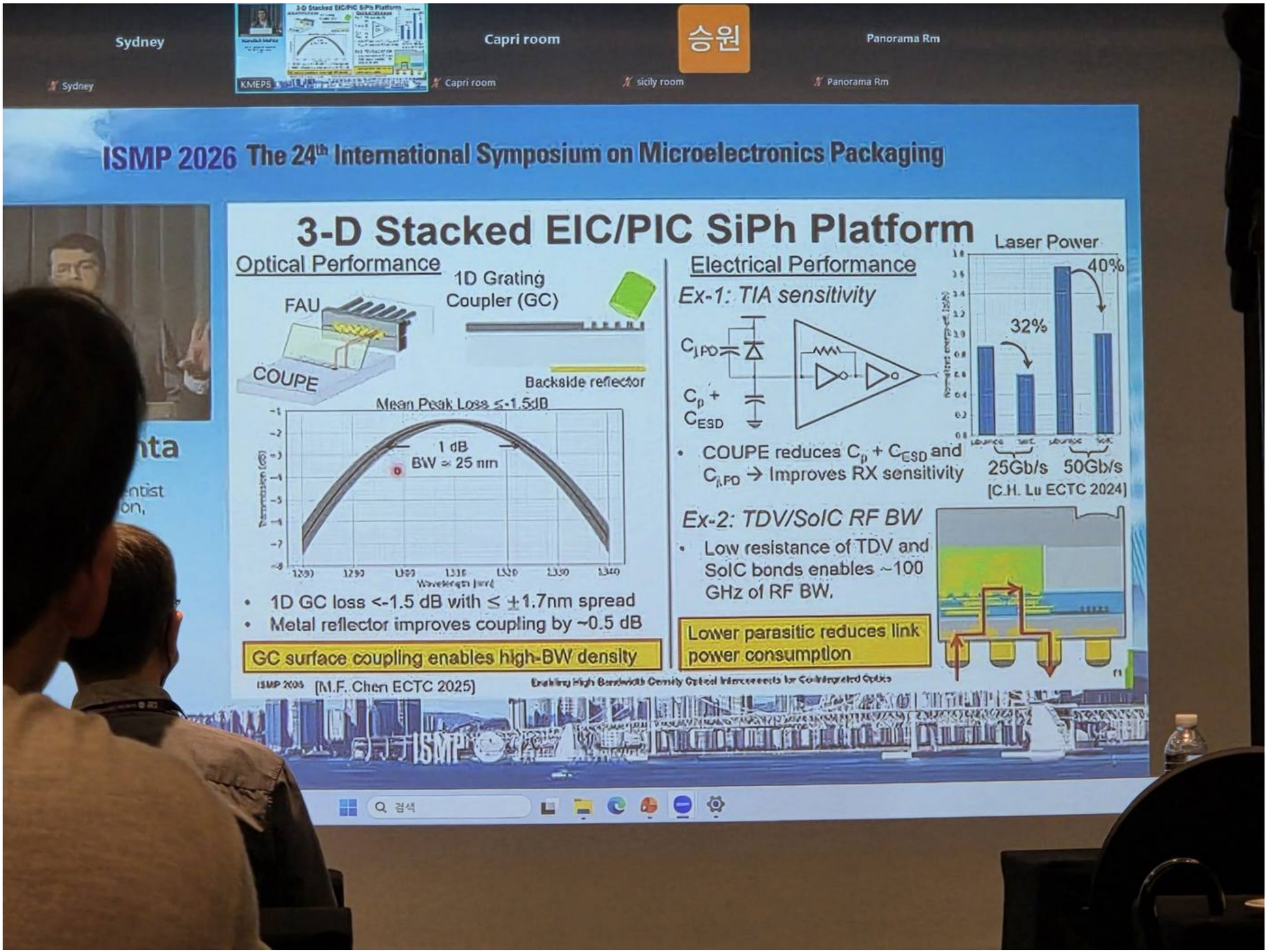

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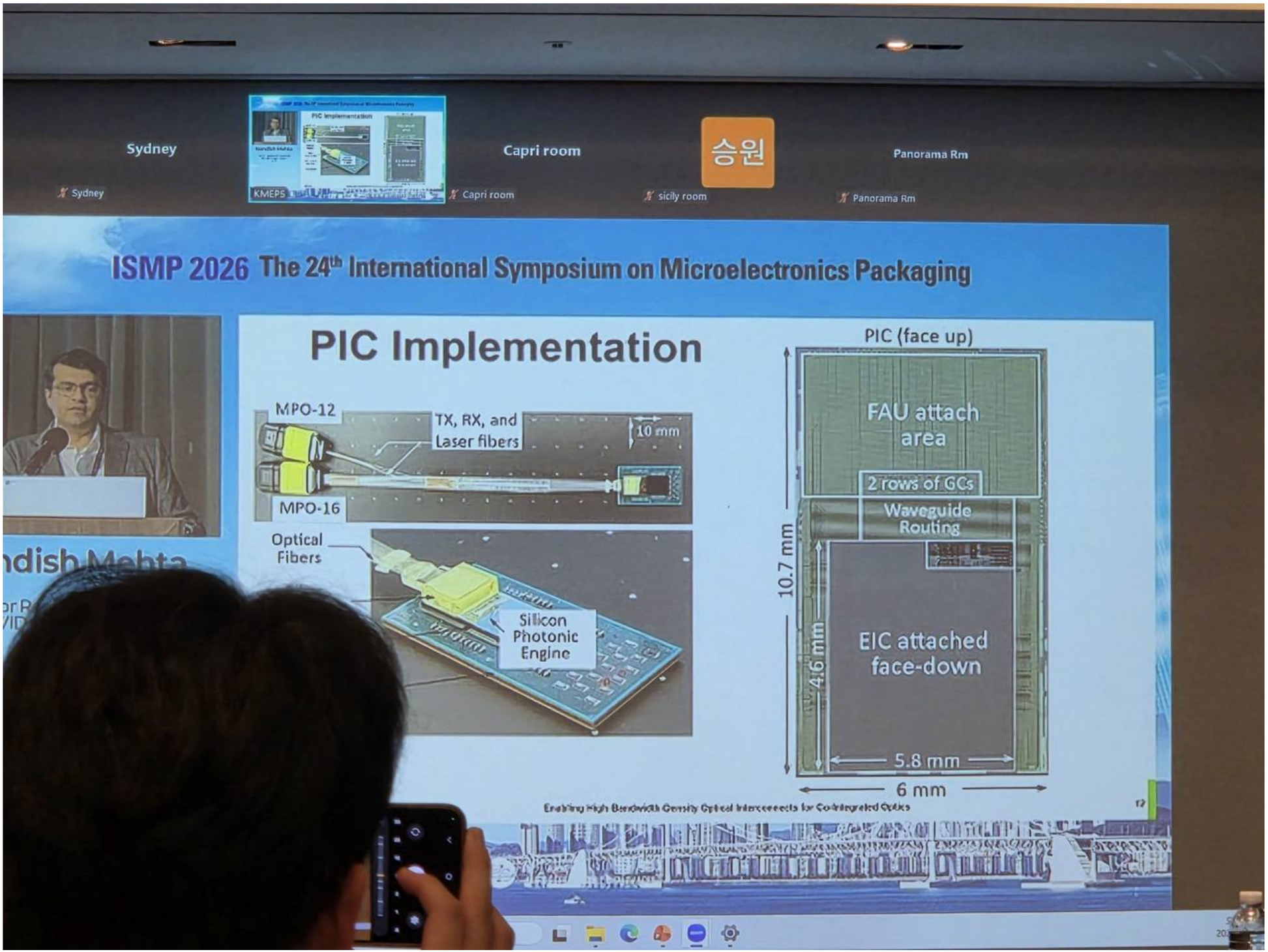

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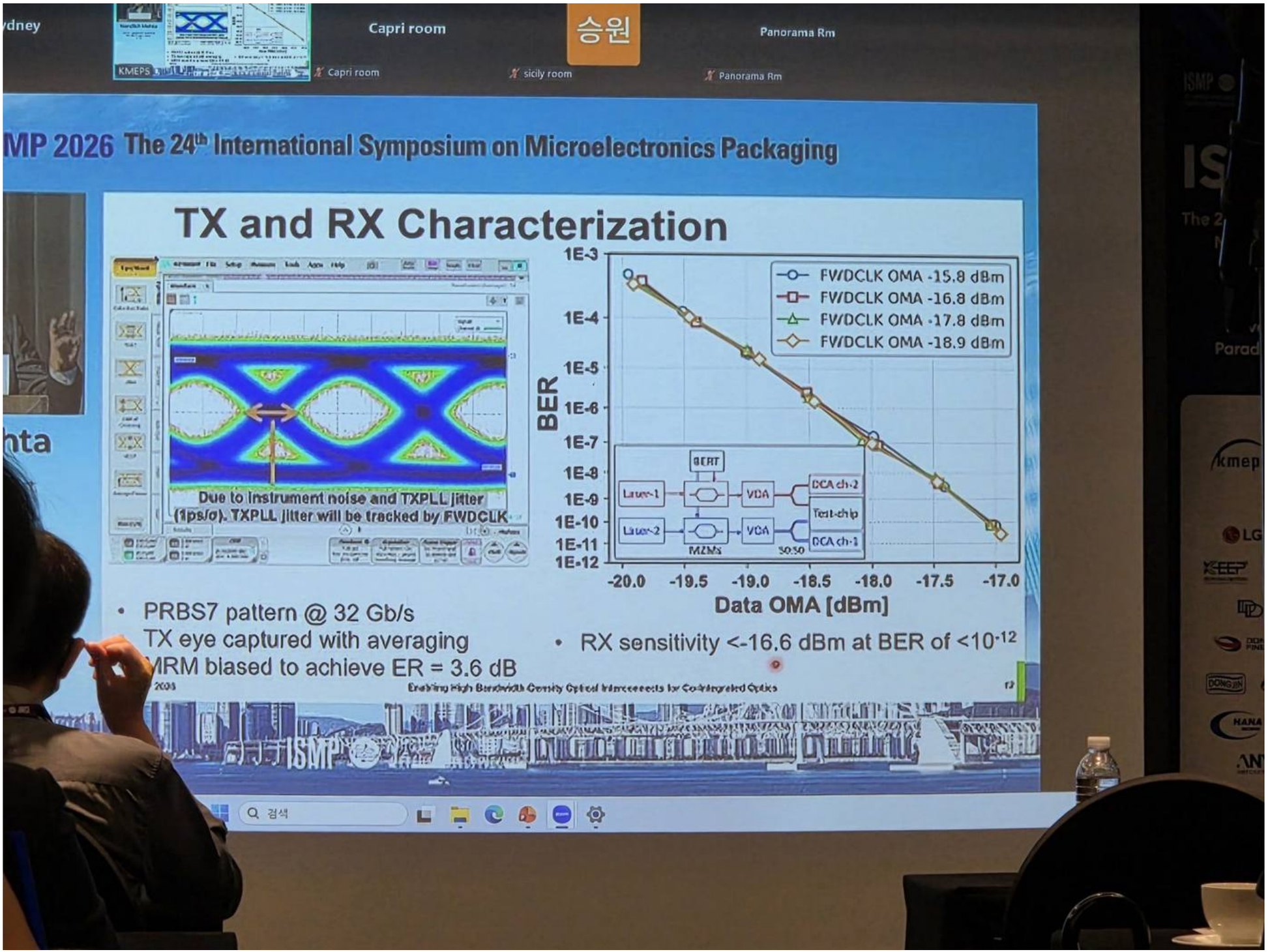

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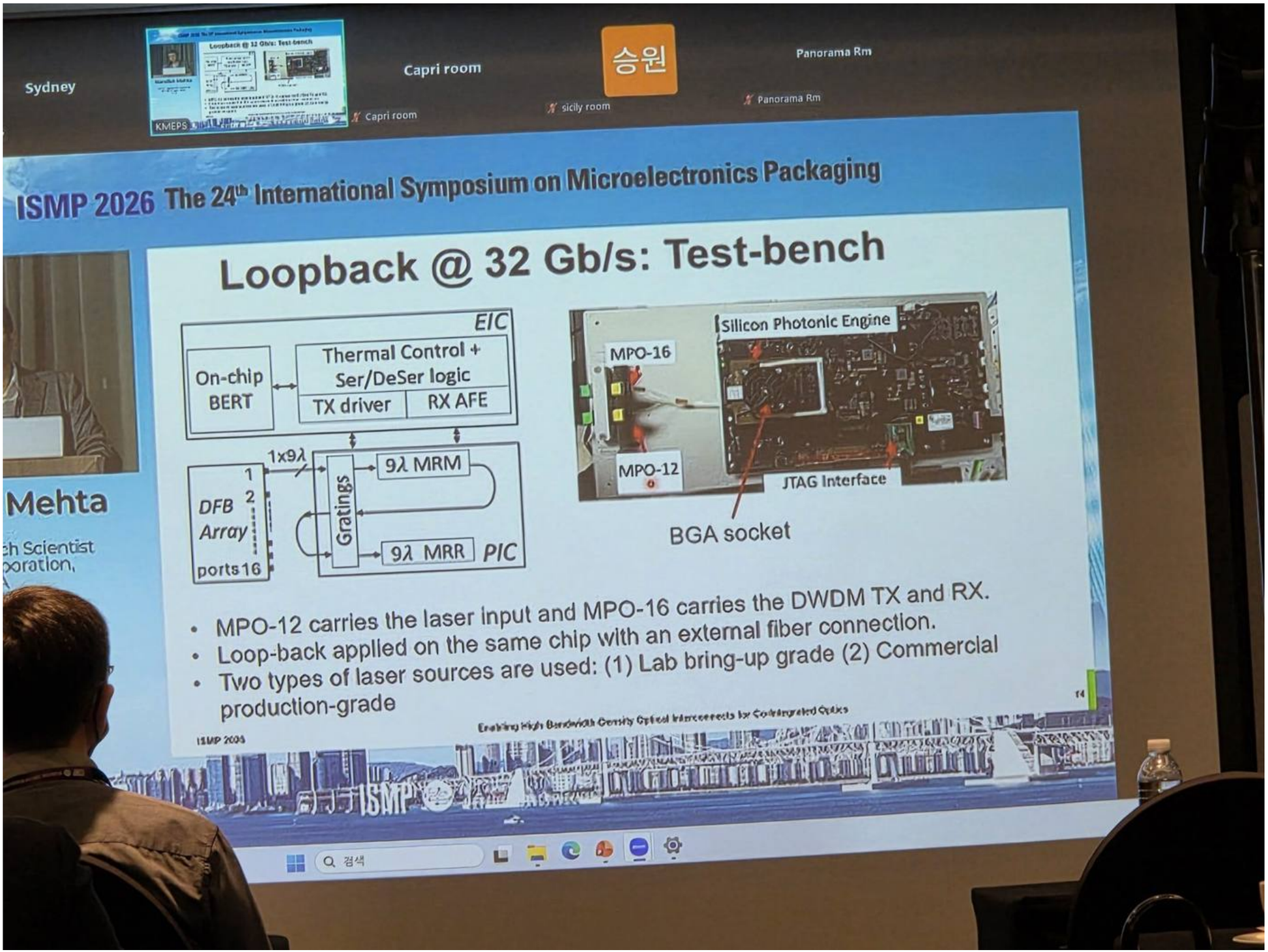

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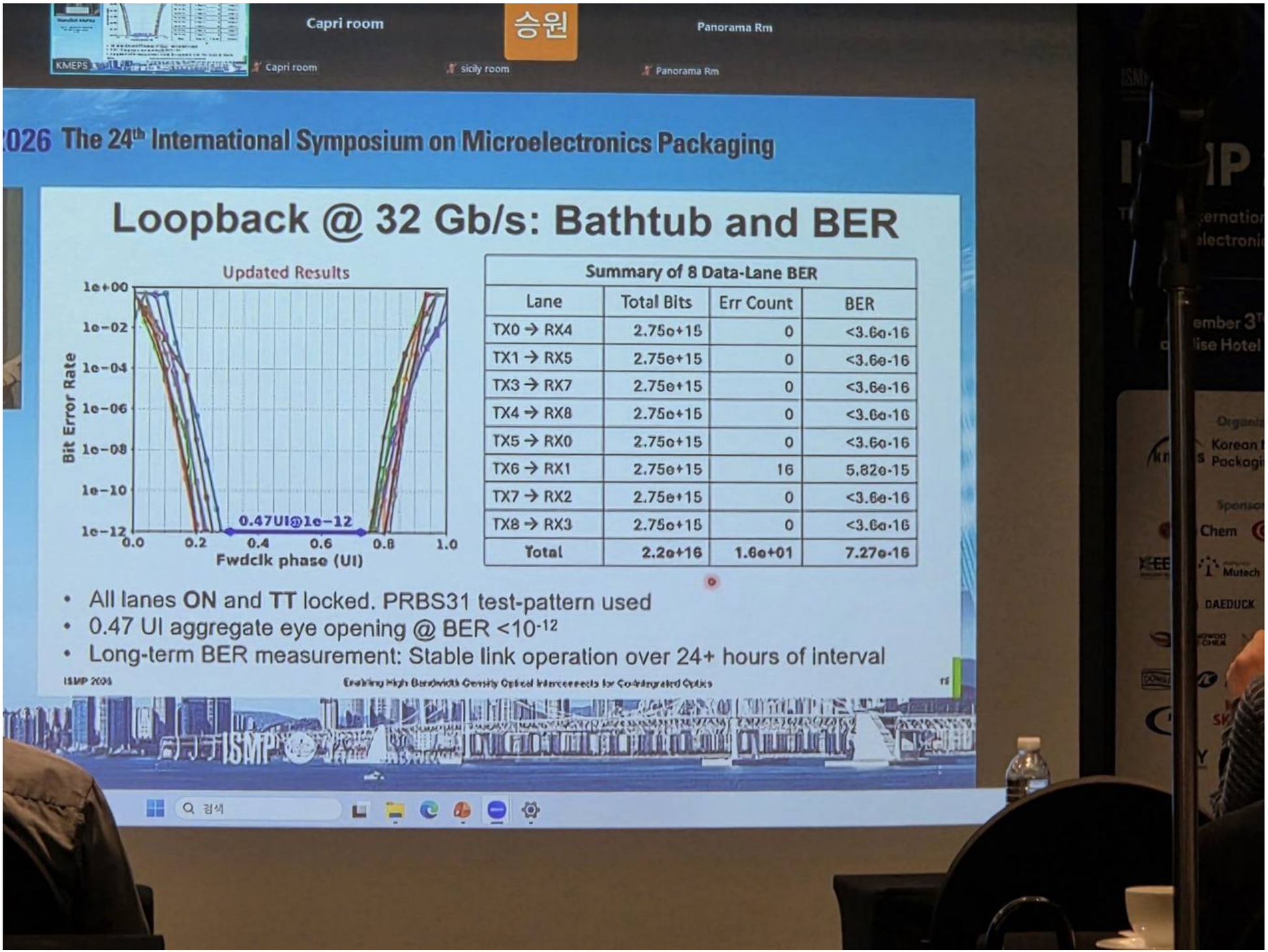

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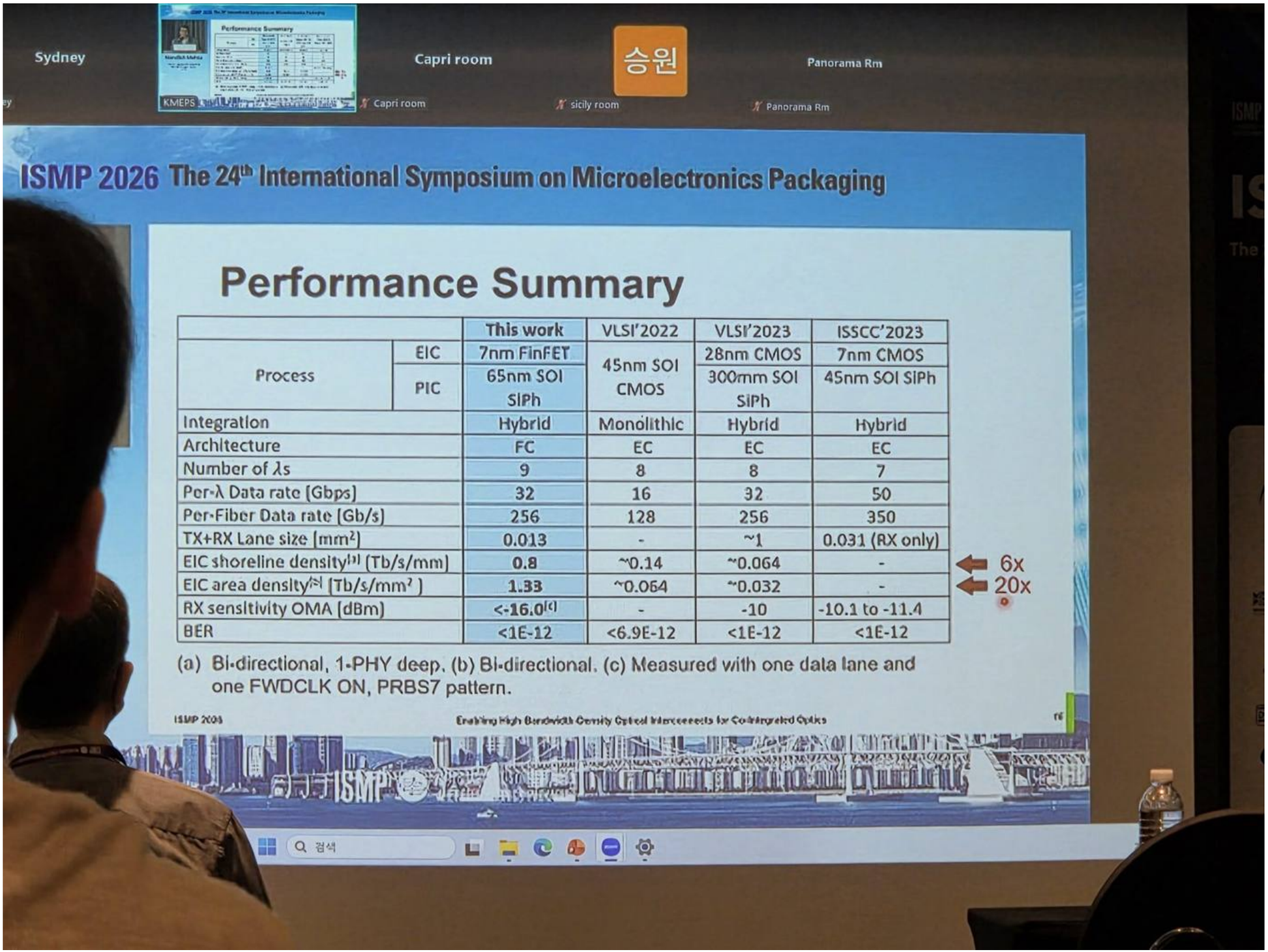

		This work	VLSI'2022	VLSI'2023	ISSCC'2023
Process	EIC	7nm FinFET	45nm SOI CMOS	28nm CMOS	7nm CMOS
	PIC	65nm SOI SiPh		300mm SOI SiPh	45nm SOI SiPh
Integration		Hybrid	Monolithic	Hybrid	Hybrid
Architecture		FC	EC	EC	EC
Number of λs		9	8	8	7
Per-λ Data rate (Gbps)		32	16	32	50
Per-Fiber Data rate (Gb/s)		256	128	256	350
TX+RX Lane size (mm²)		0.013	-	~1	0.031 (RX only)
EIC shoreline density[b] (Tb/s/mm)		0.8	~0.14	~0.064	-
EIC area density[b] (Tb/s/mm²)		1.33	~0.064	~0.032	-
RX sensitivity OMA (dBm)		<-16.0[c]	-	-10	-10.1 to -11.4
BER		<1E-12	<6.9E-12	<1E-12	<1E-12

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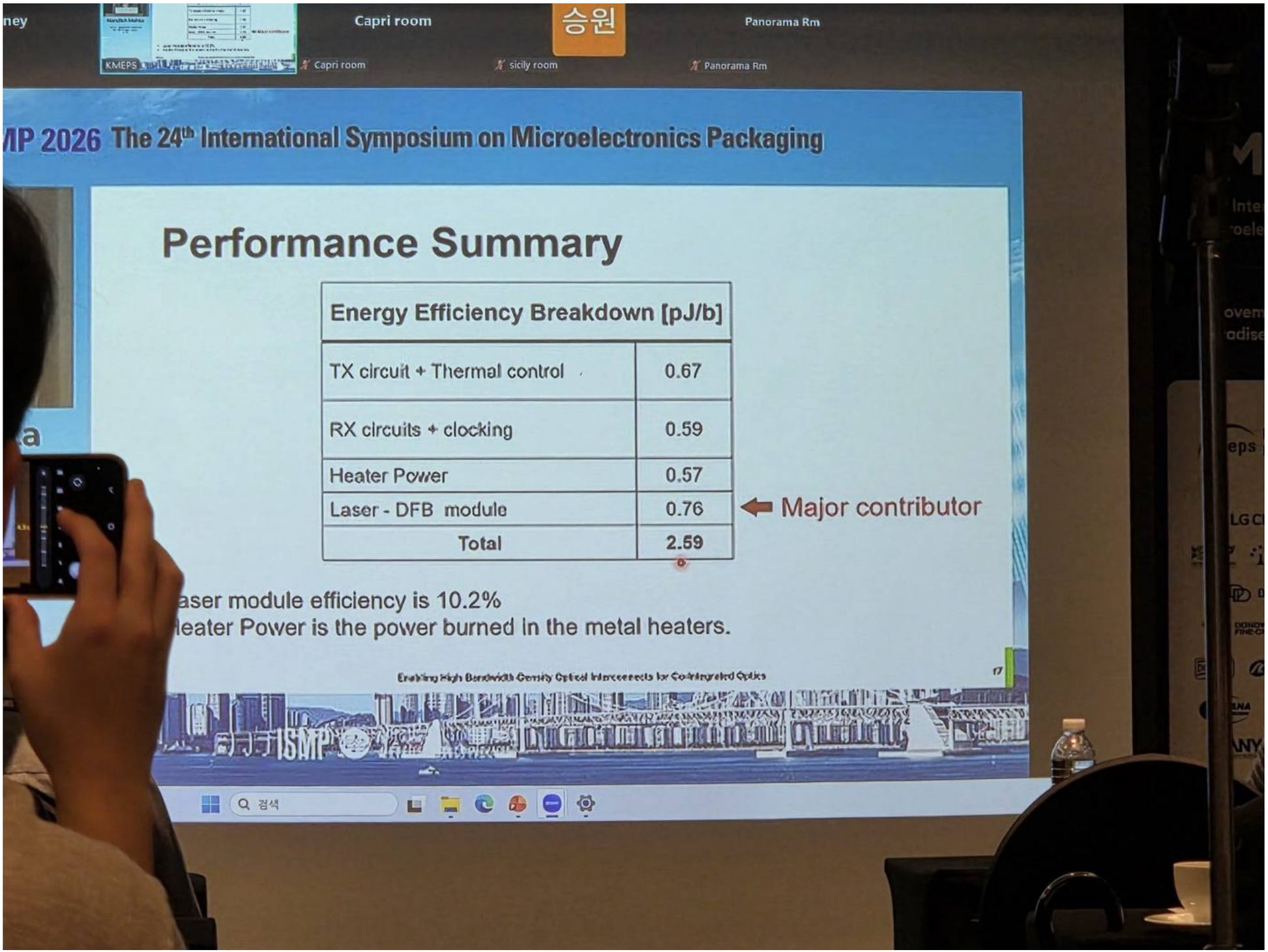

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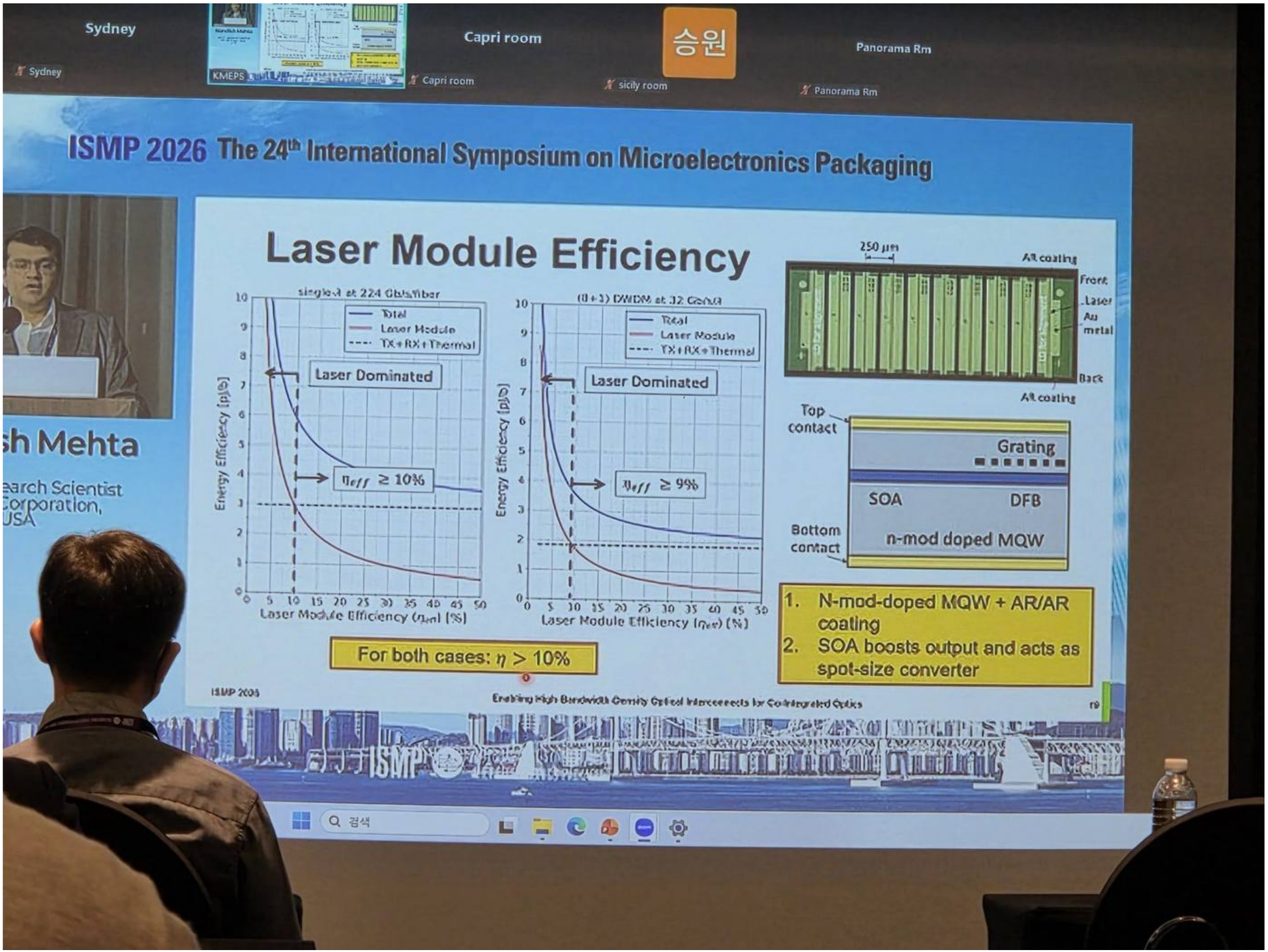

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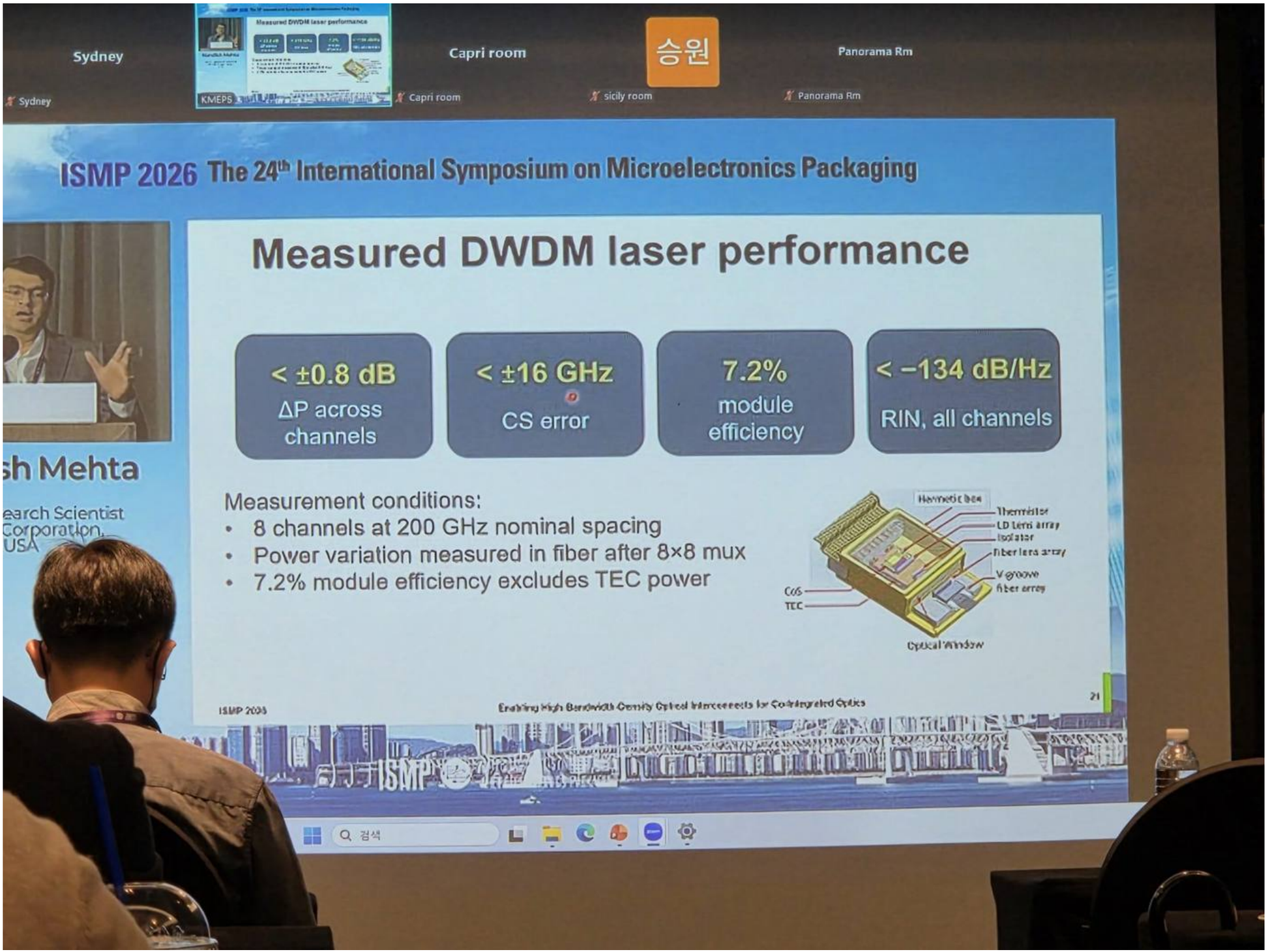

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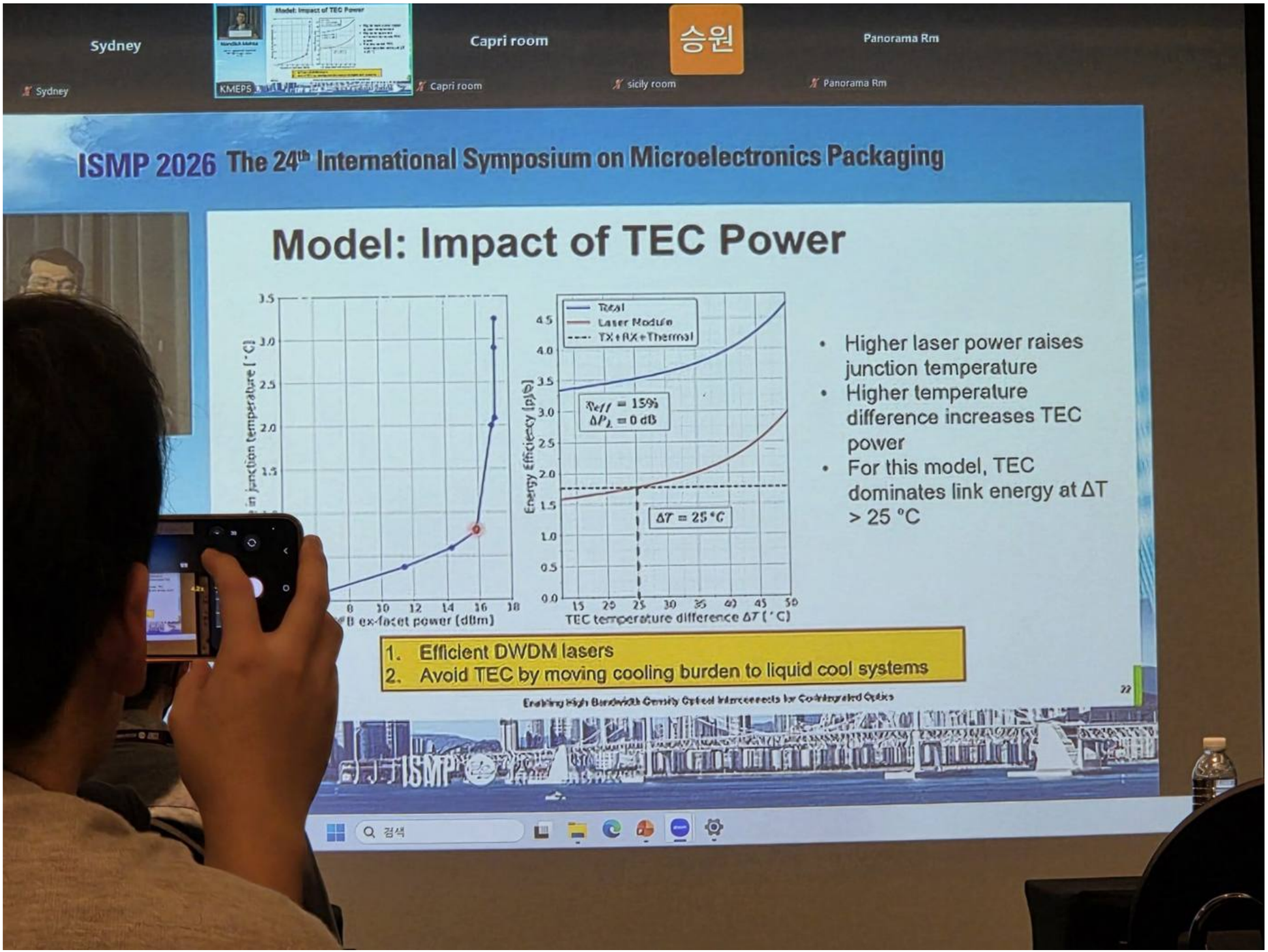

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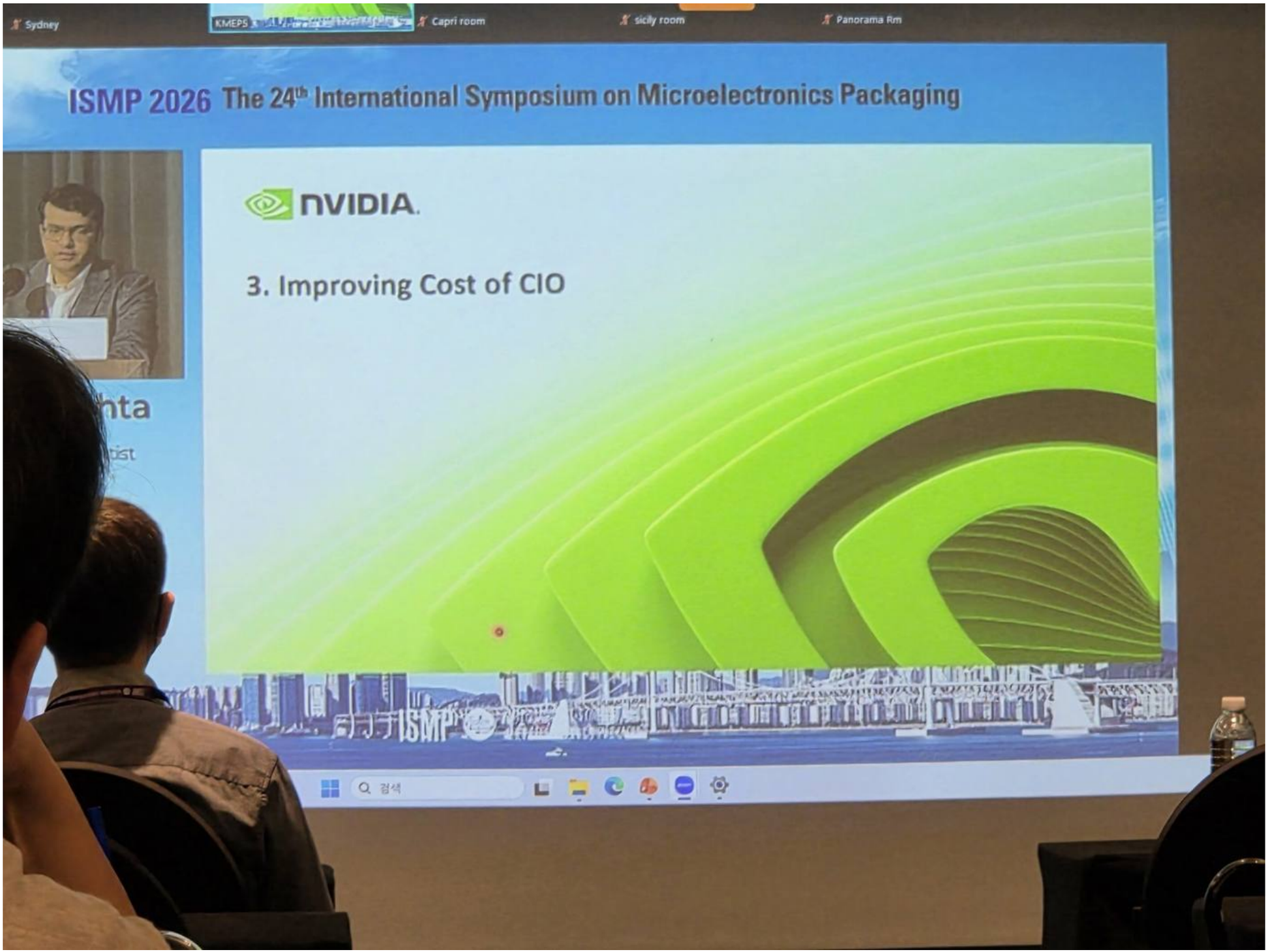

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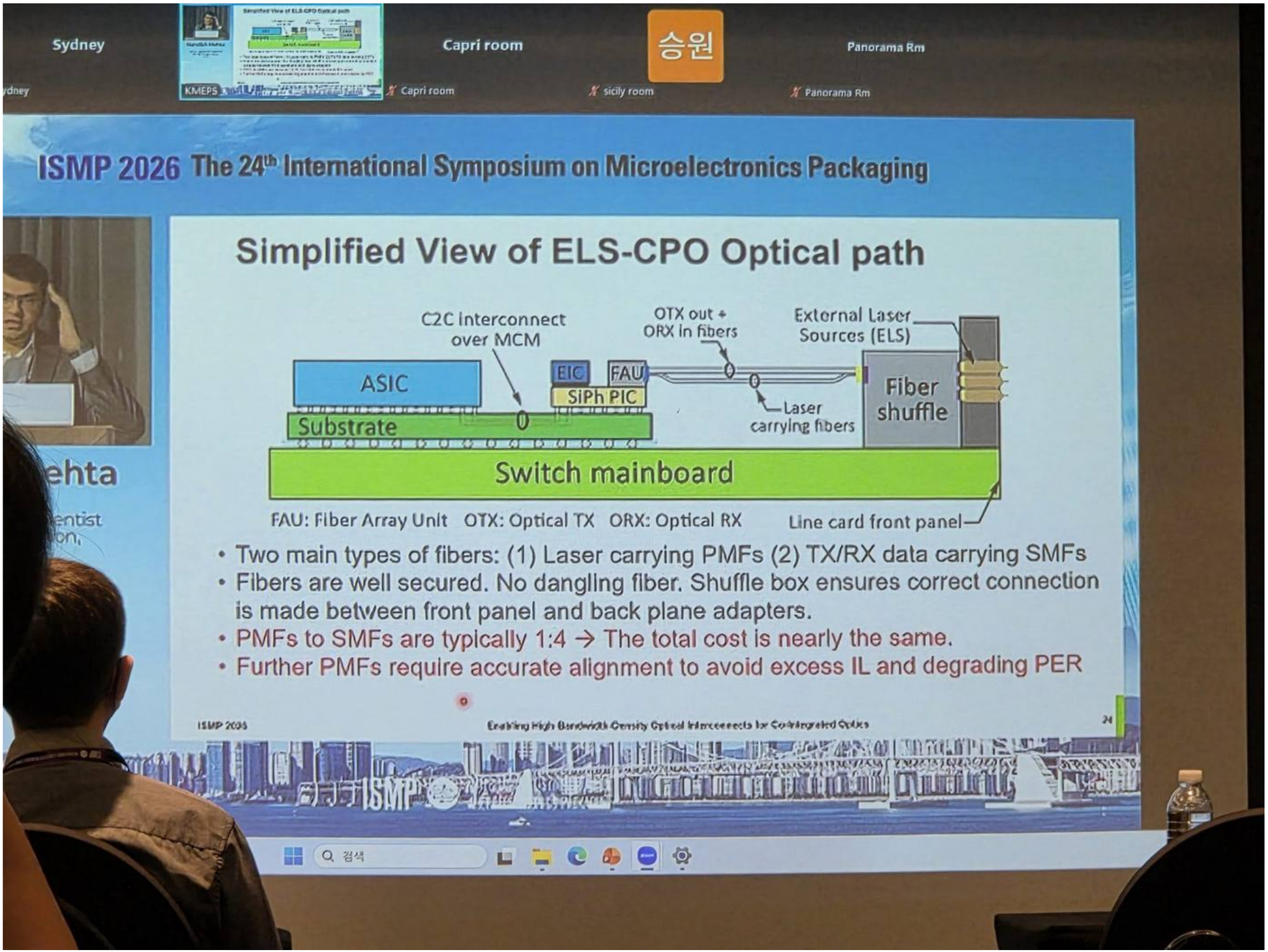

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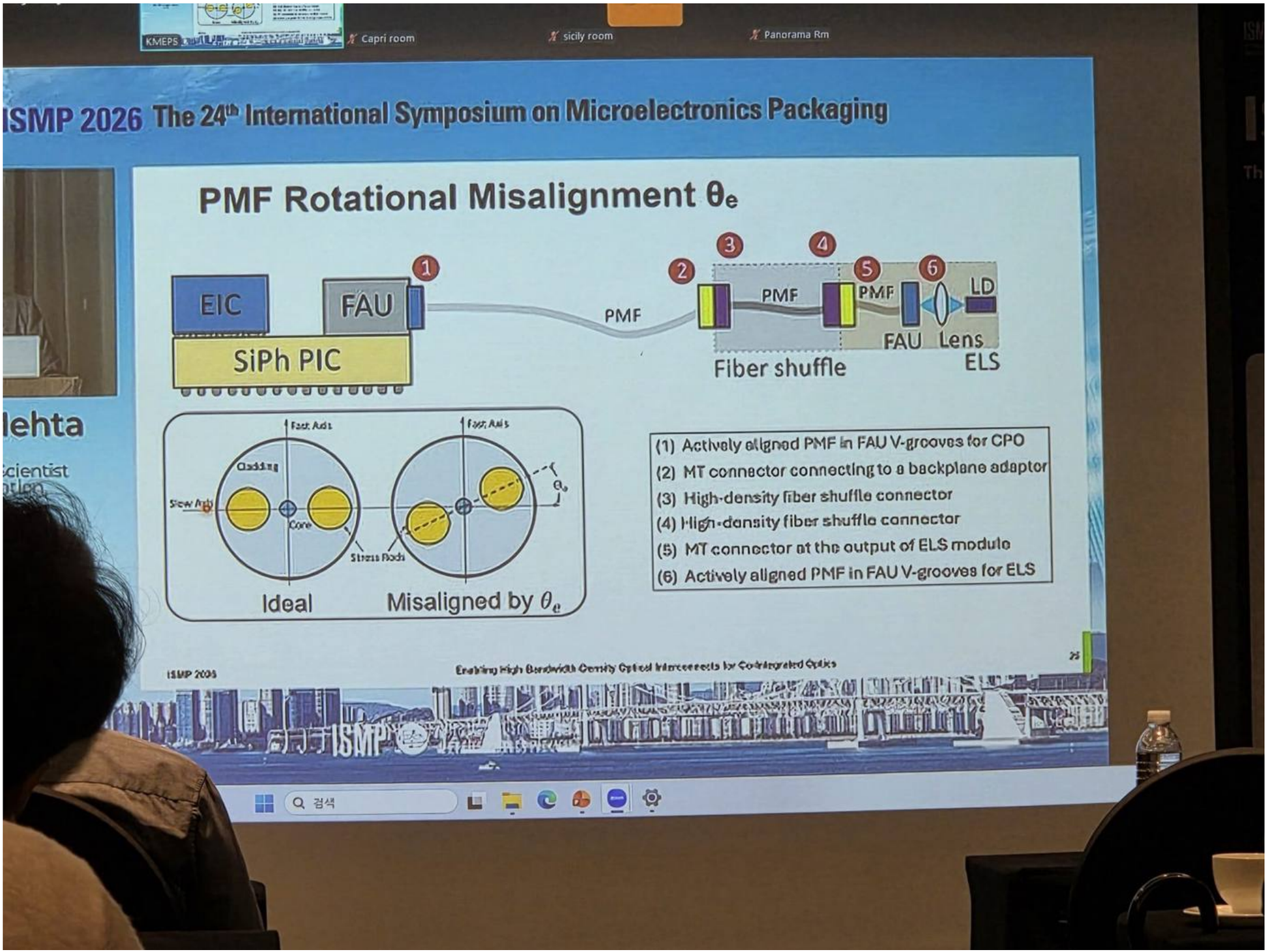

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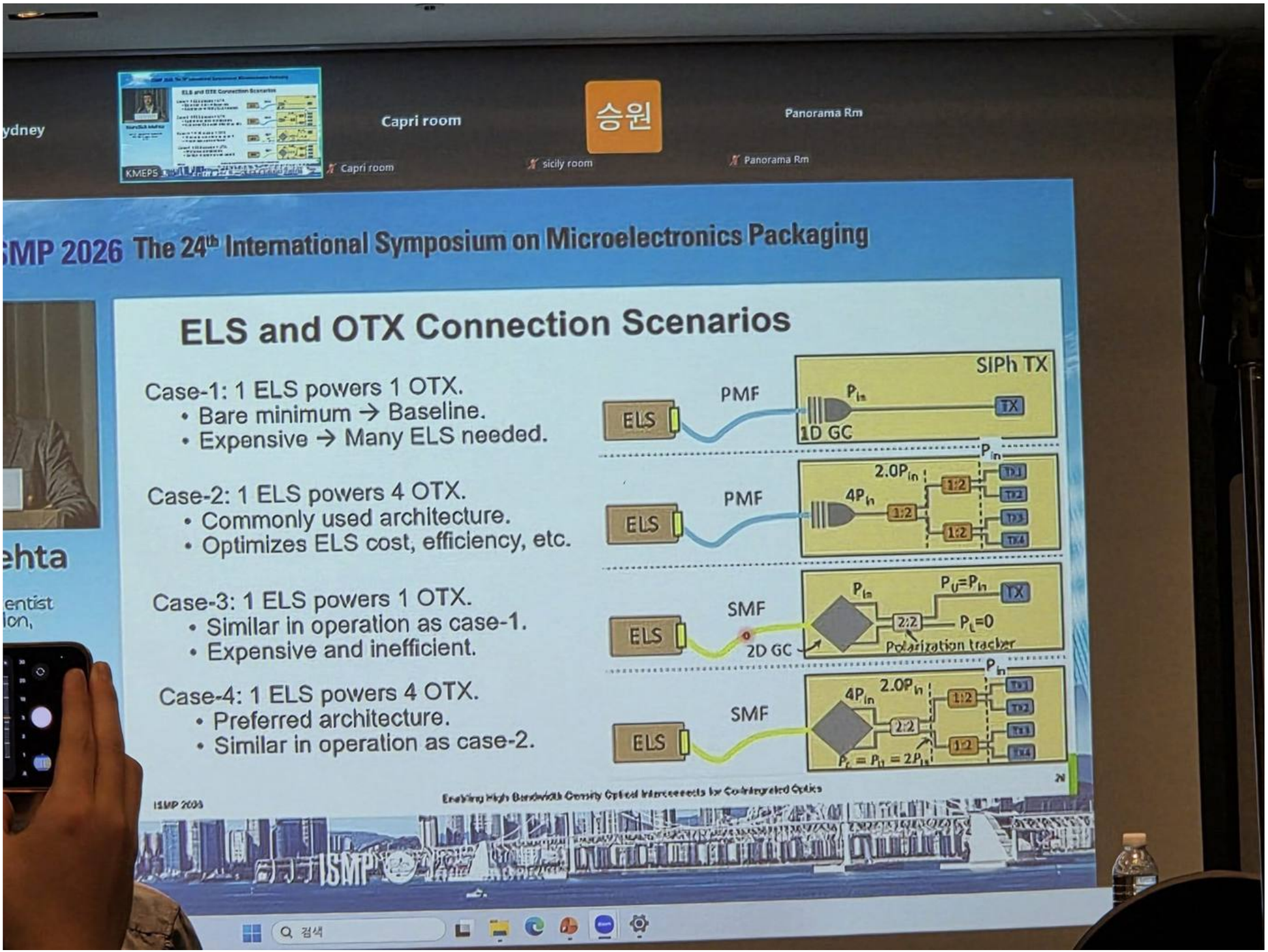

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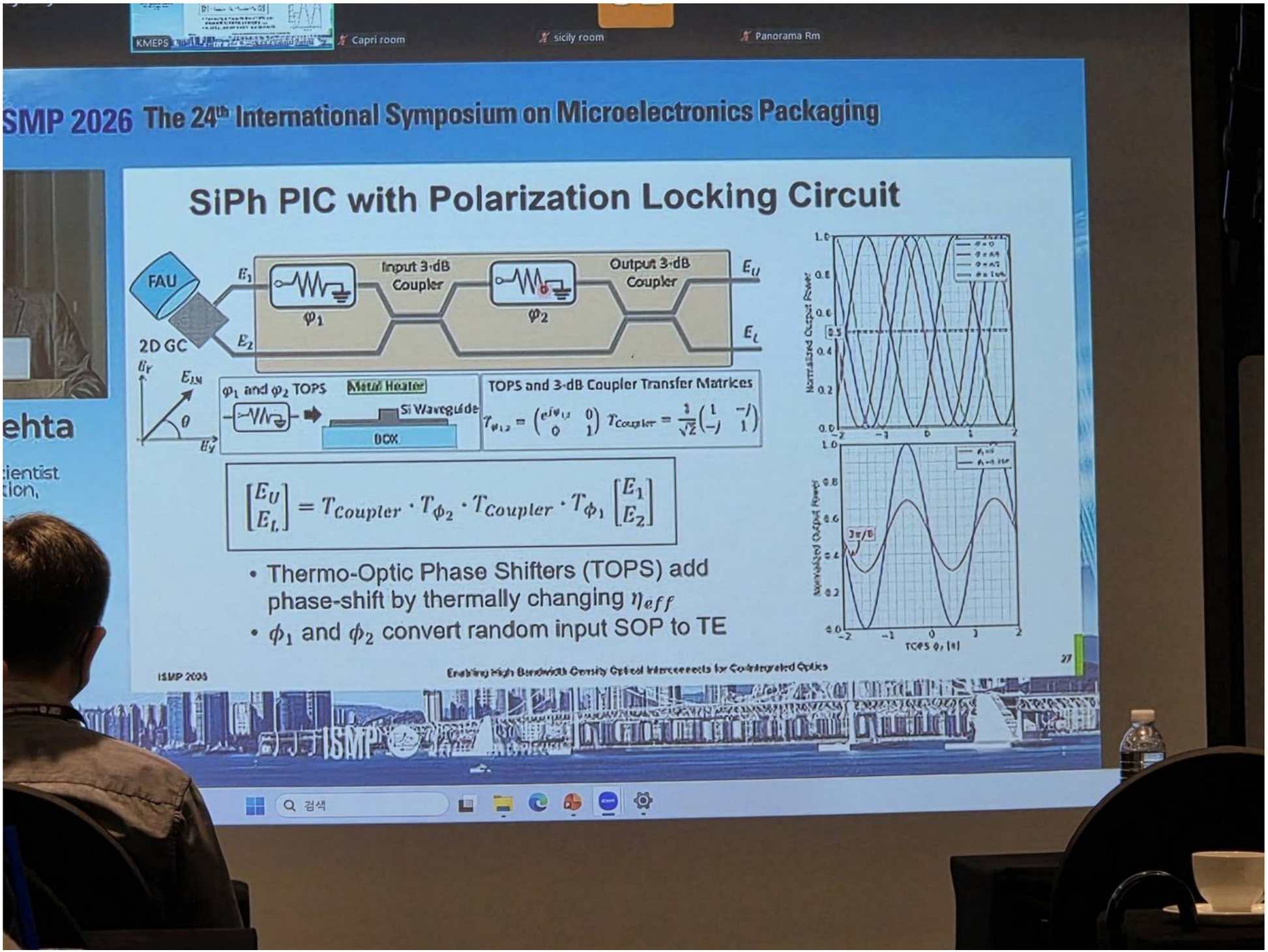

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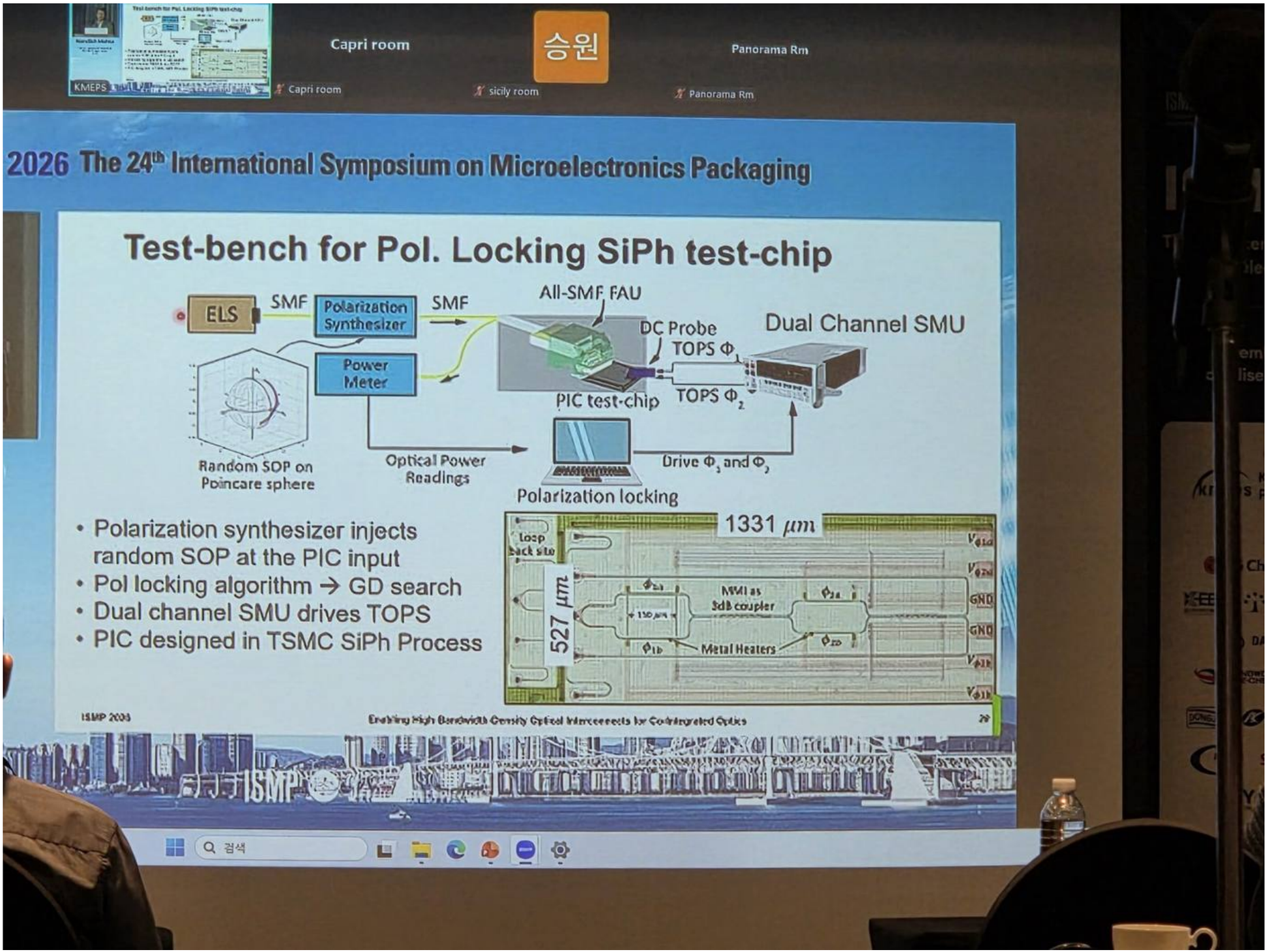

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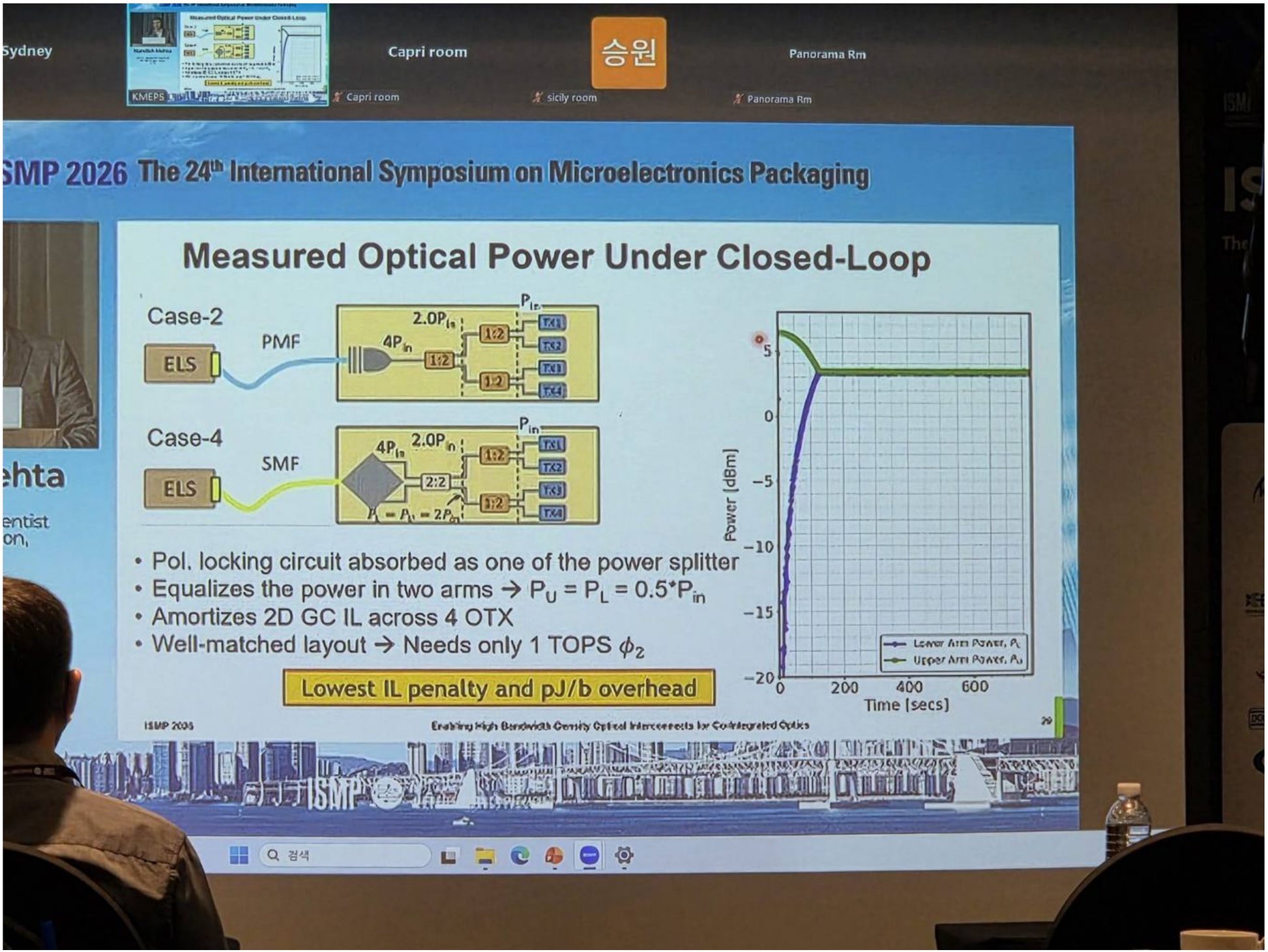

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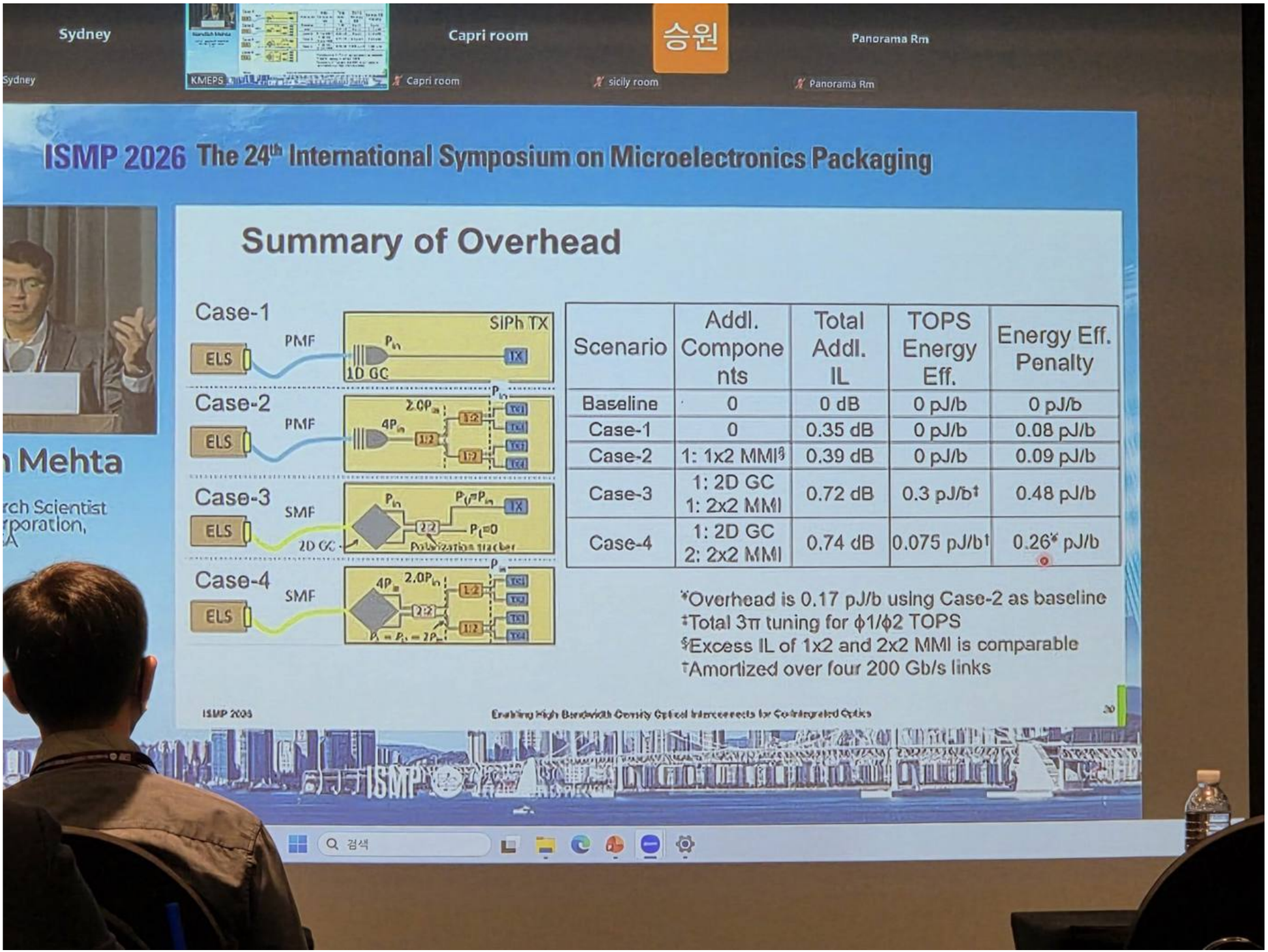

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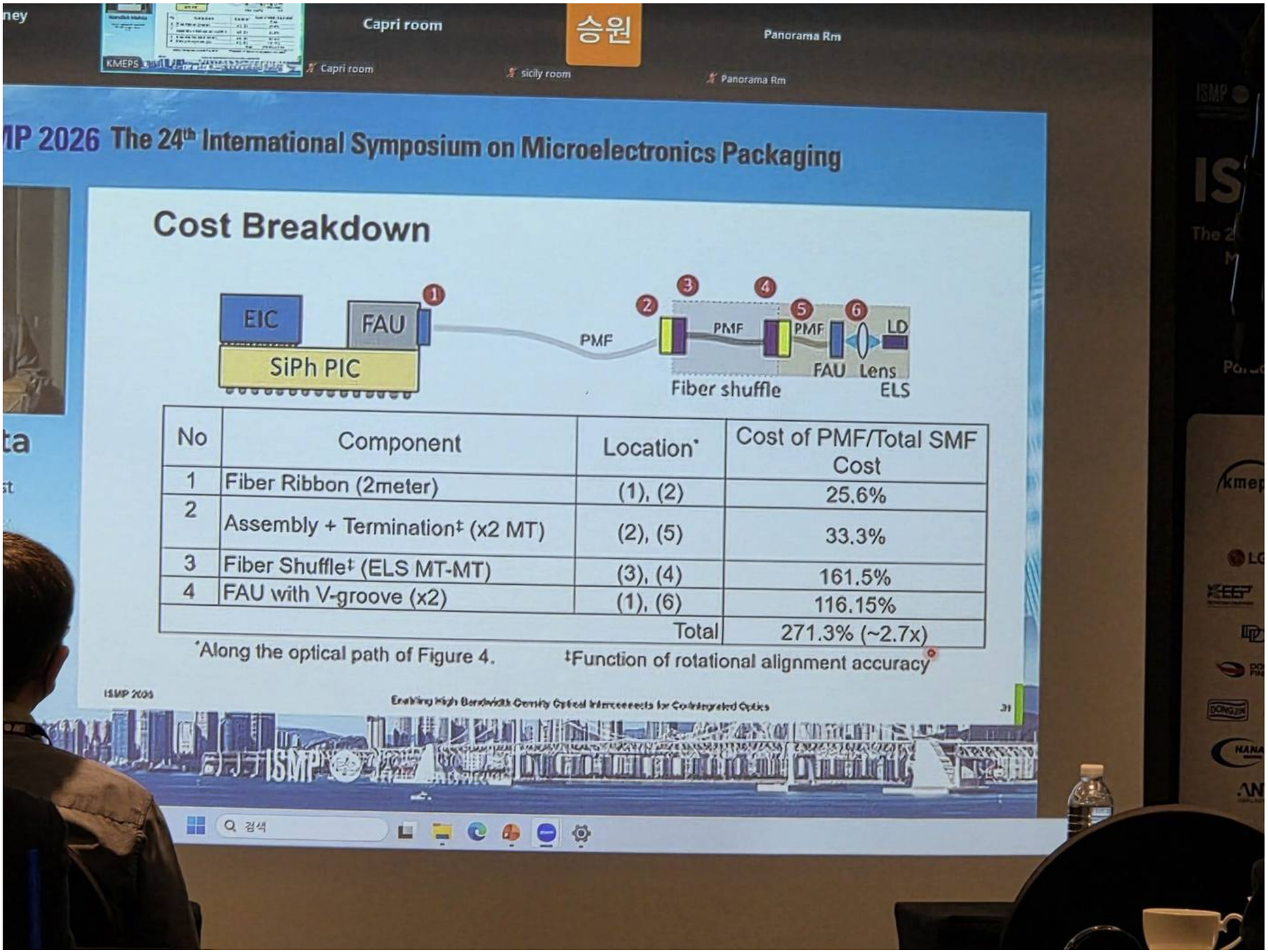

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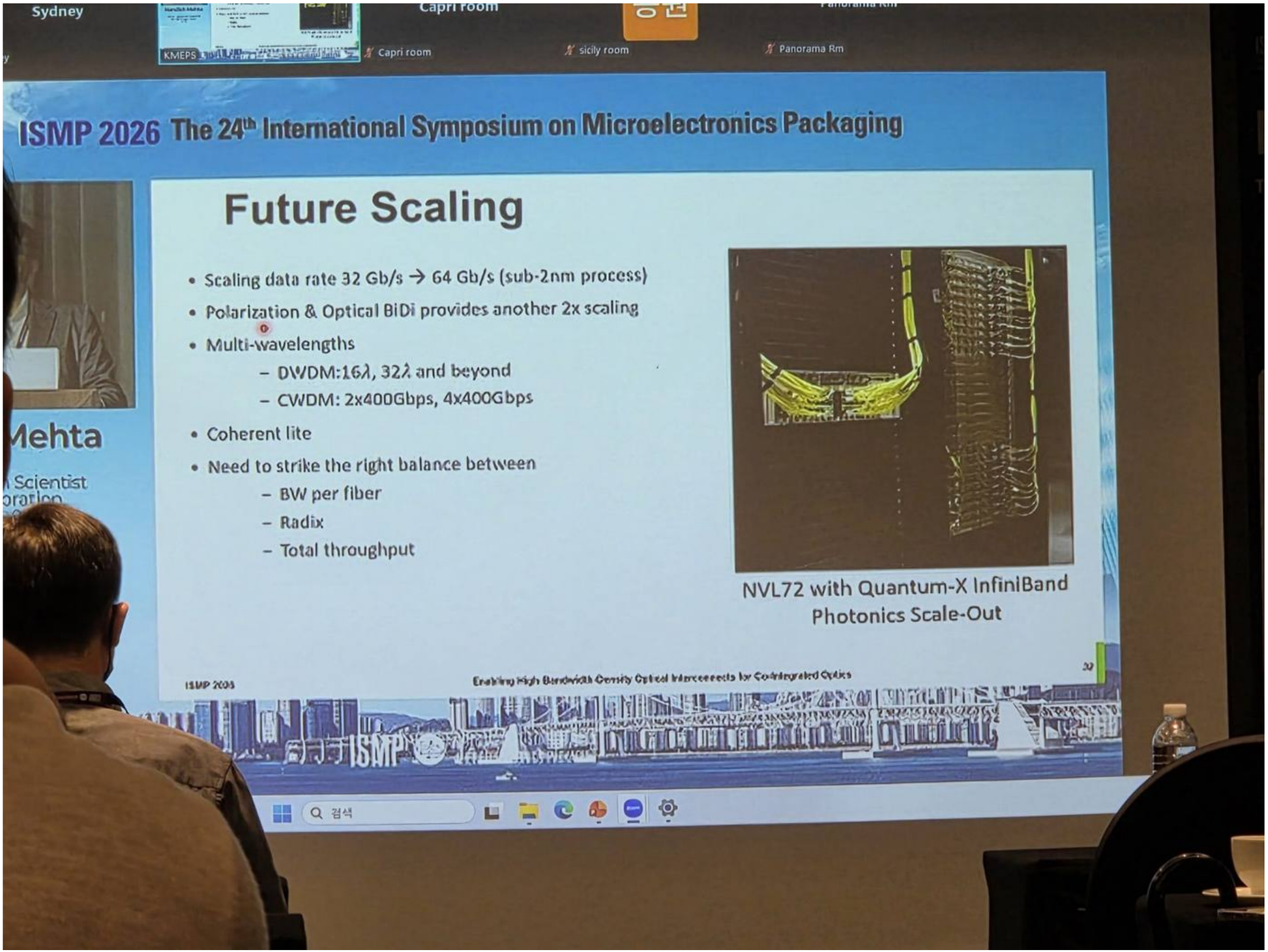

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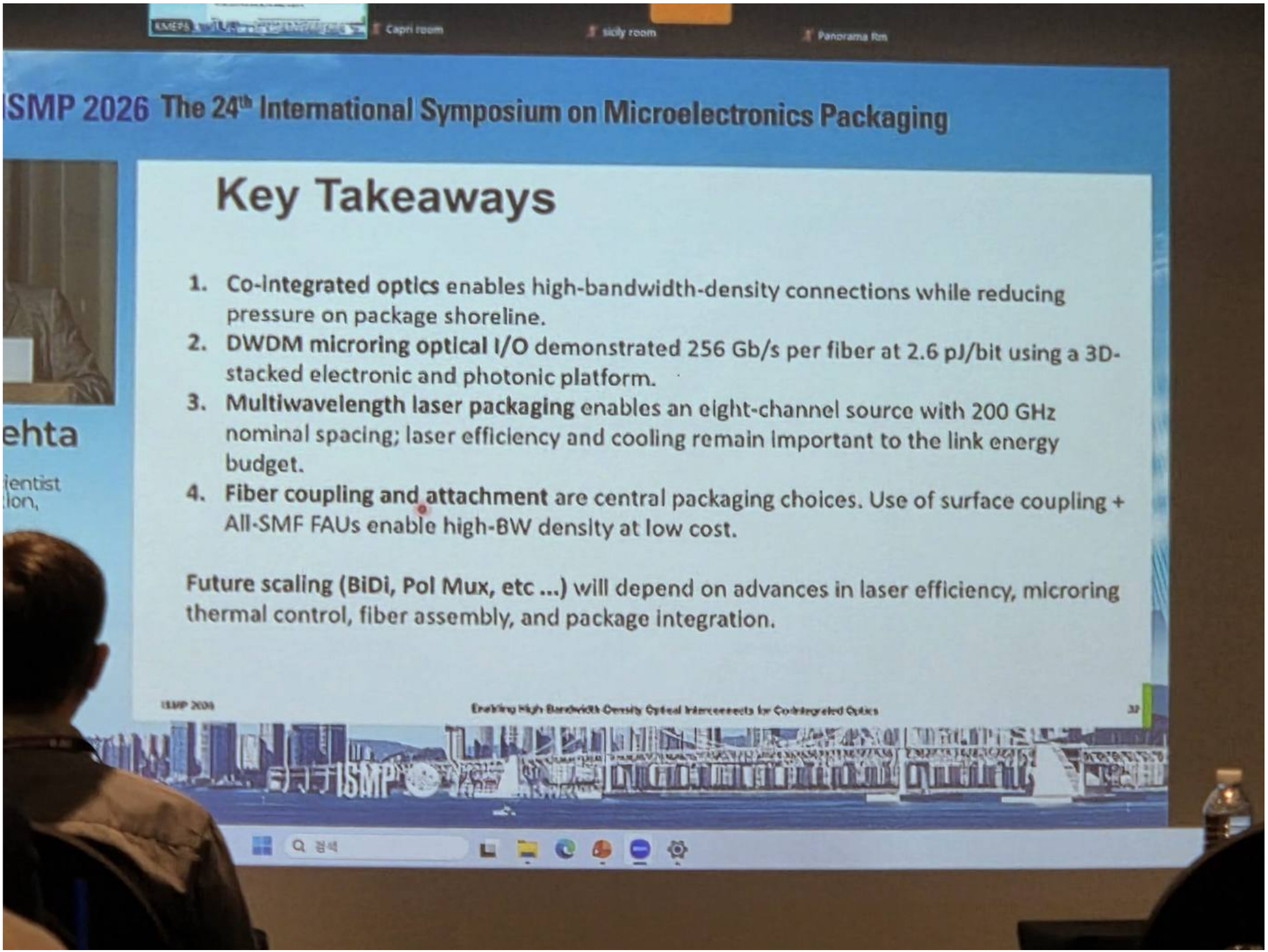

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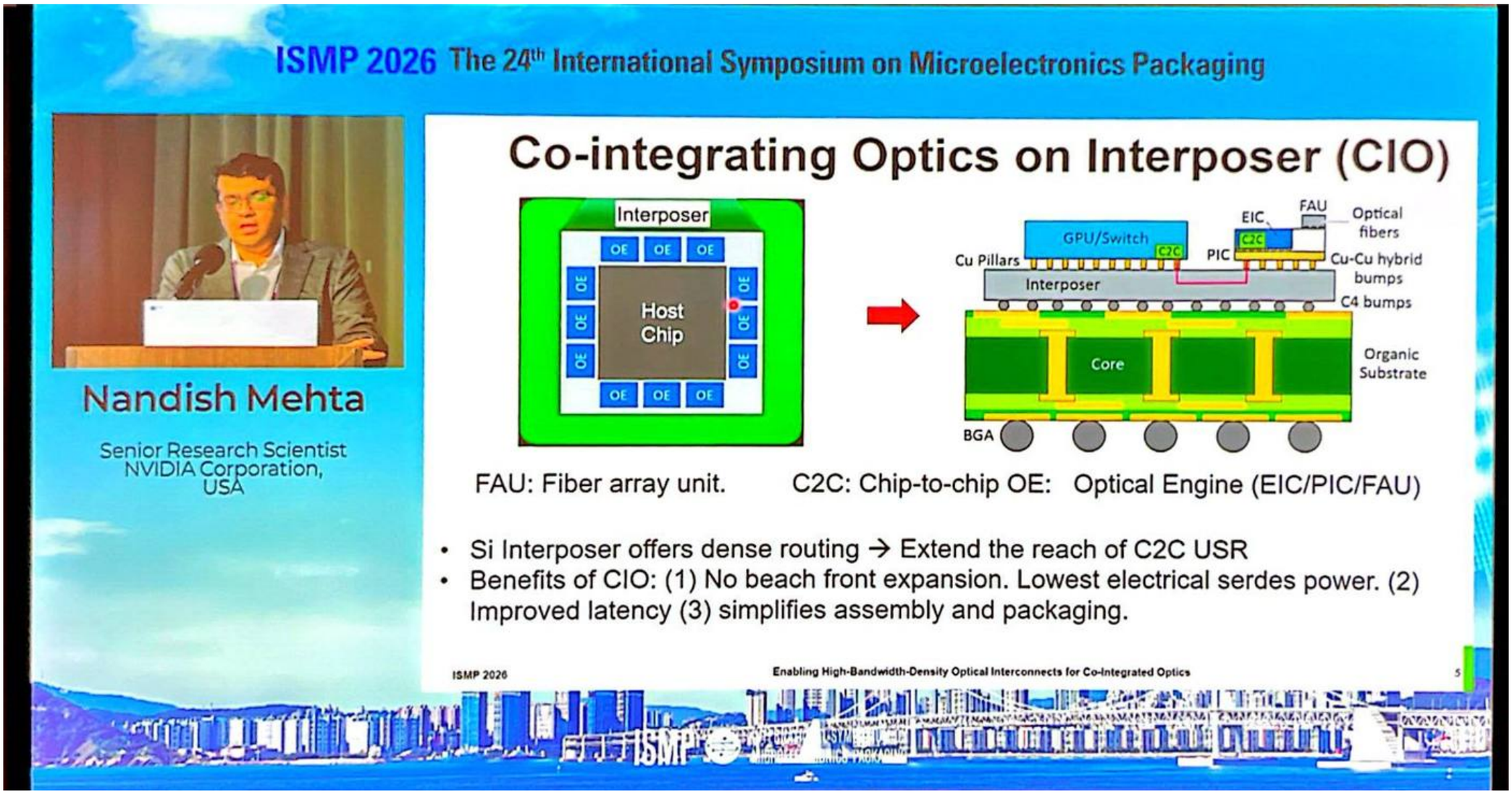

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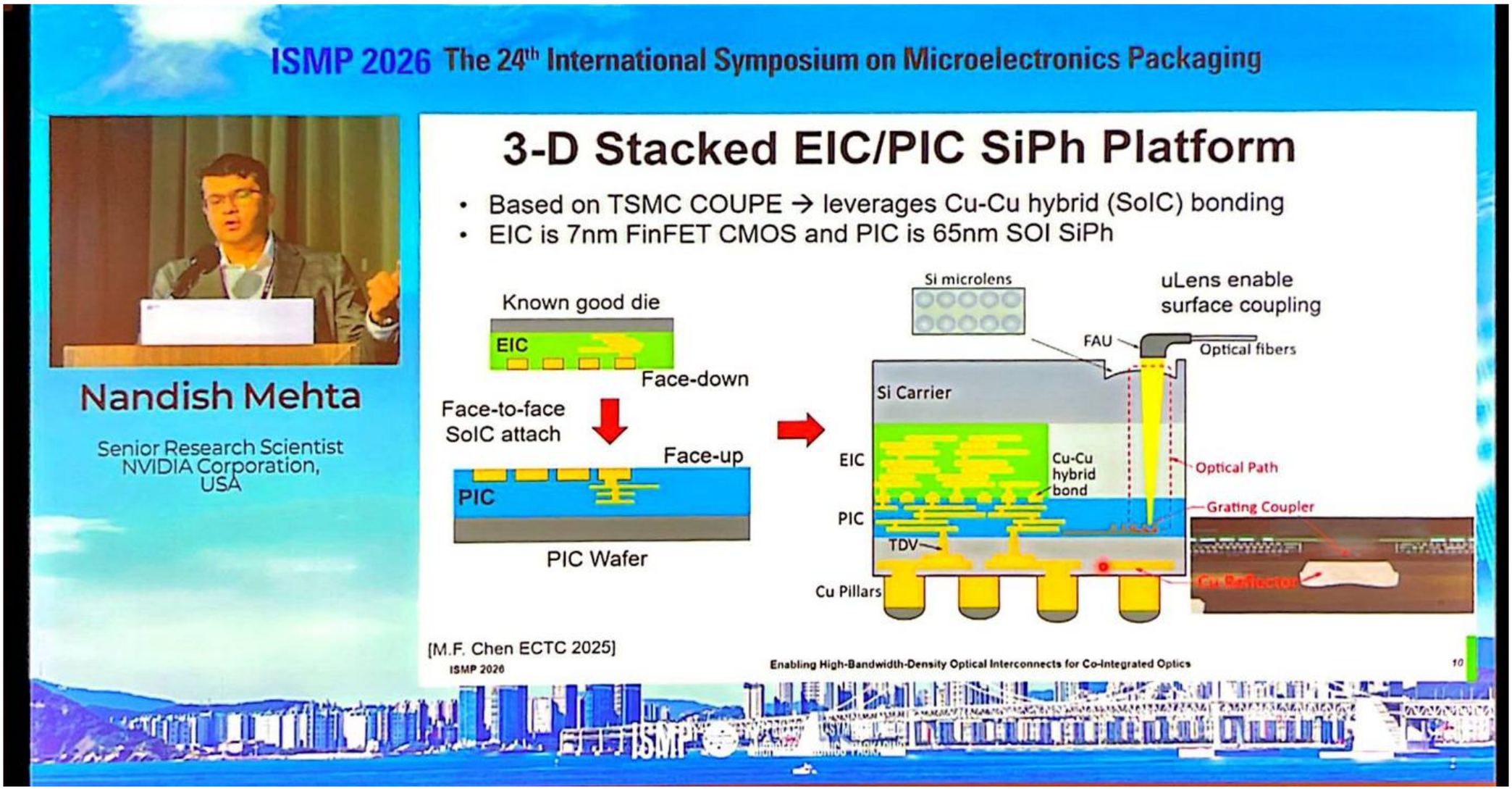

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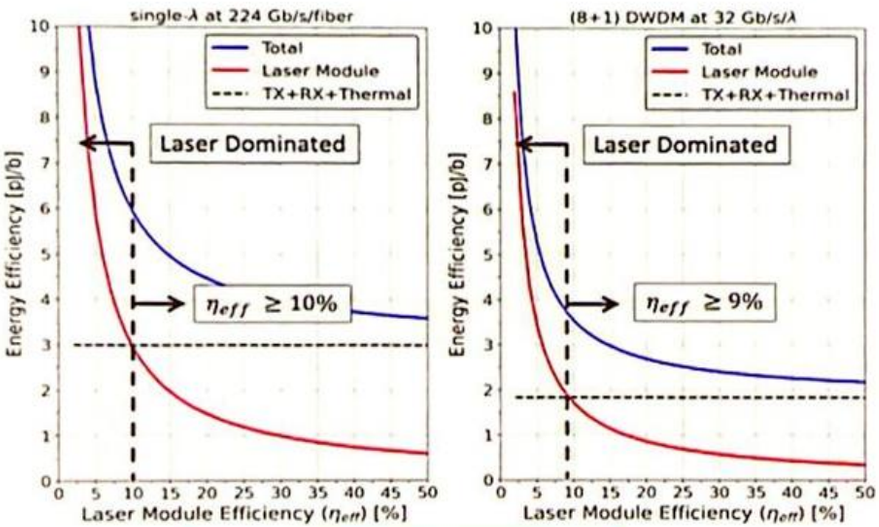
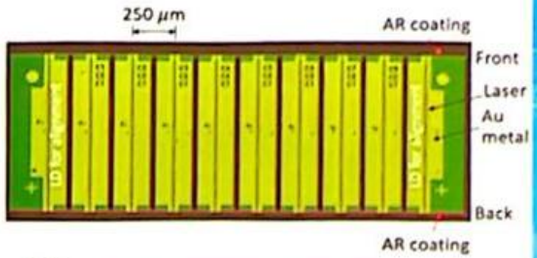
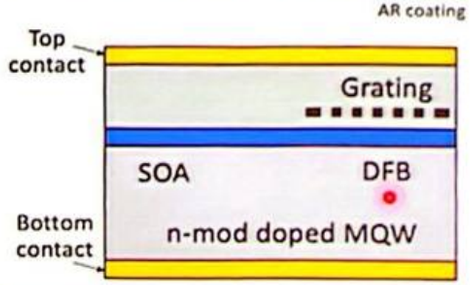
Nandish Mehta

Senior Research Scientist
NVIDIA Corporation,
USA

Laser Module Efficiency







For both cases: $\eta > 10\%$

1. N-mod-doped MQW + AR/AR coating
2. SOA boosts output and acts as spot-size converter

ISMP 2026 Enabling High-Bandwidth-Density Optical Interconnects for Co-Integrated Optics 19

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Nandish Mehta

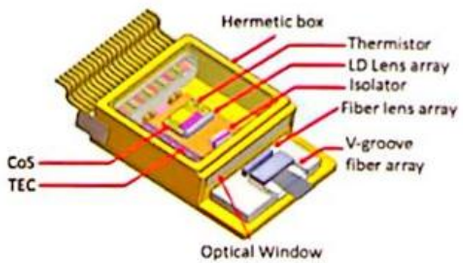
Senior Research Scientist
NVIDIA Corporation,
USA

Measured DWDM laser performance

< ±0.8 dB	< ±16 GHz	7.2%	< −134 dB/Hz
ΔP across channels	CS error	module efficiency	RIN, all channels

Measurement conditions:

- 8 channels at 200 GHz nominal spacing
- Power variation measured in fiber after 8×8 mux
- 7.2% module efficiency excludes TEC power



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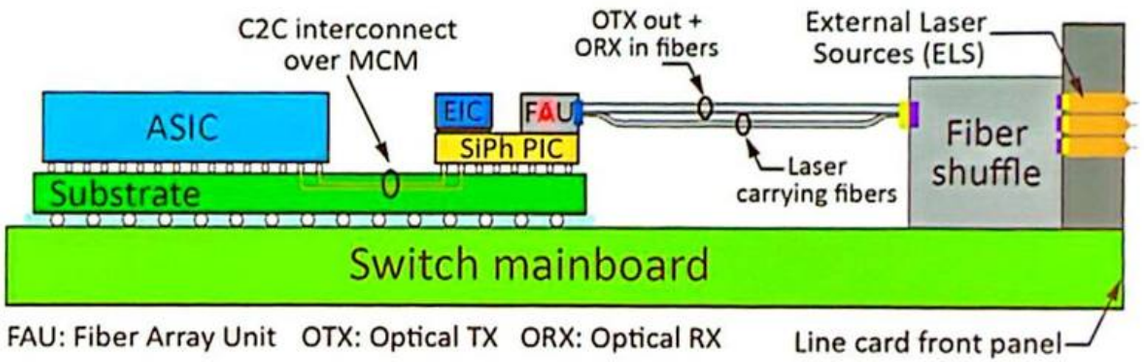
Nandish Mehta

Senior Research Scientist
NVIDIA Corporation,
USA

Simplified View of ELS-CPO Optical path



- Two main types of fibers: (1) Laser carrying PMFs (2) TX/RX data carrying SMFs
- Fibers are well secured. No dangling fiber. Shuffle box ensures correct connection is made between front panel and back plane adapters.
- PMFs to SMFs are typically 1:4 → The total cost is nearly the same.
- Further PMFs require accurate alignment to avoid excess IL and degrading PER

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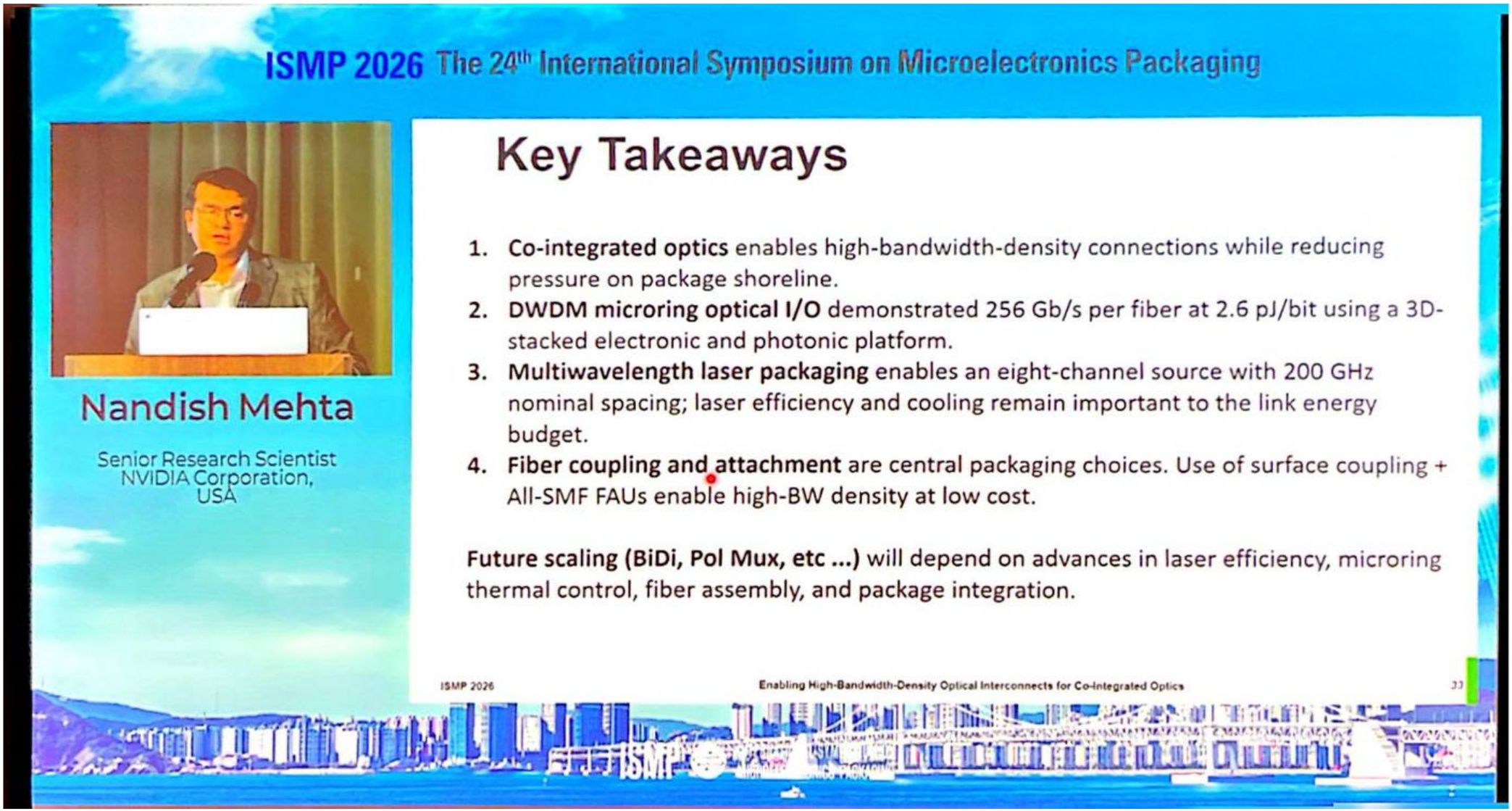

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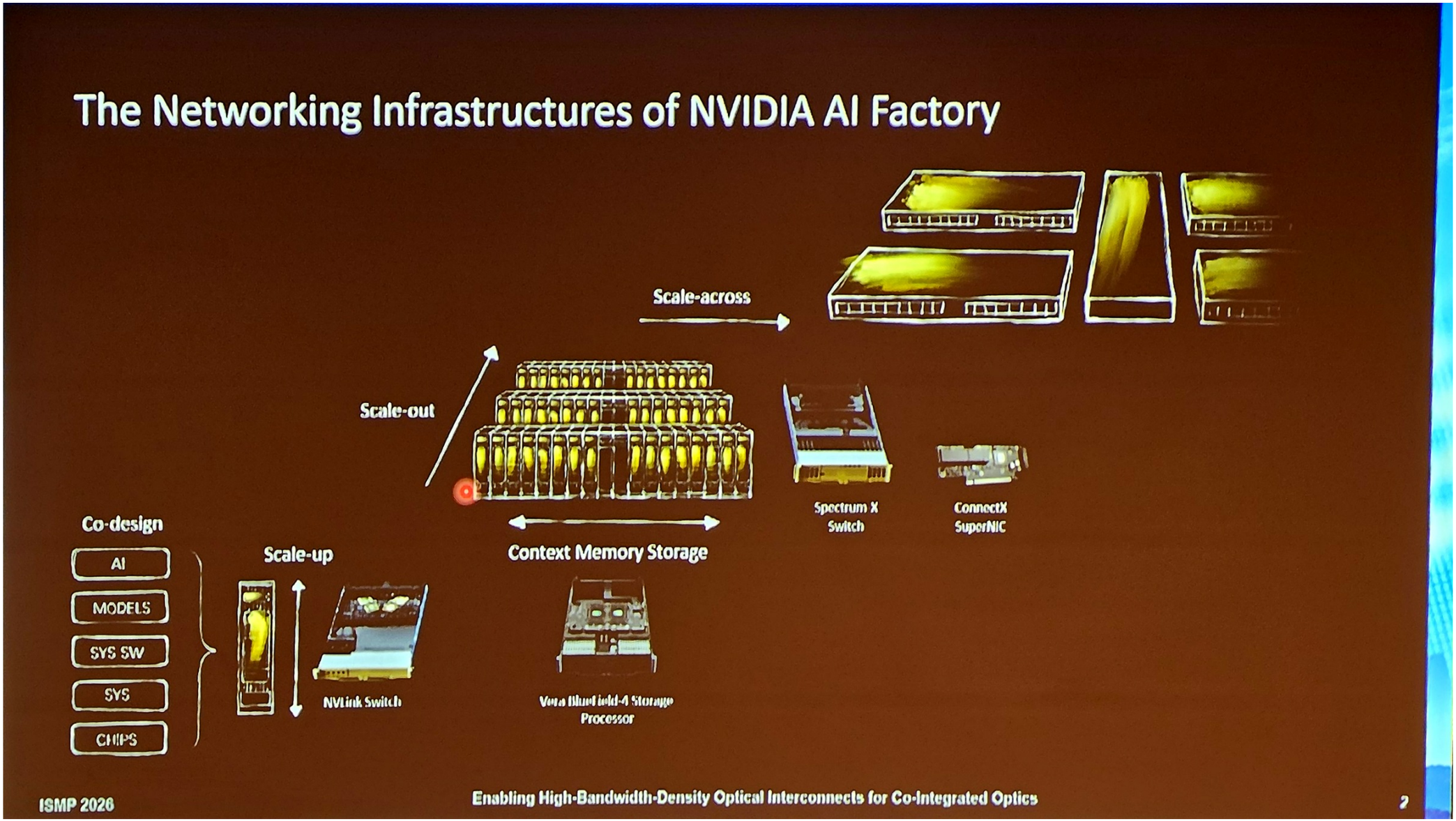

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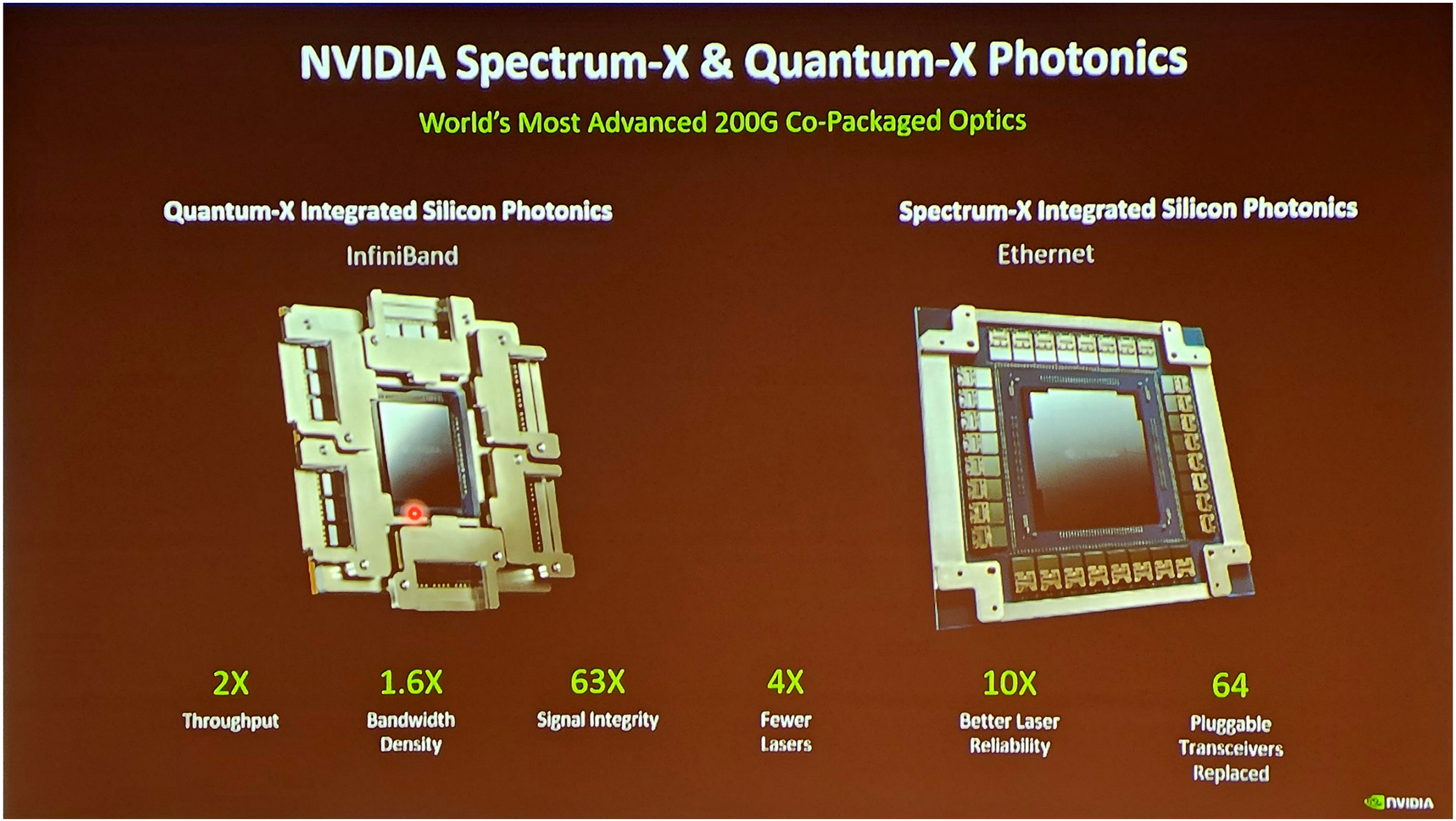

T19 인터포저 위 광학 집적과 링크 효율

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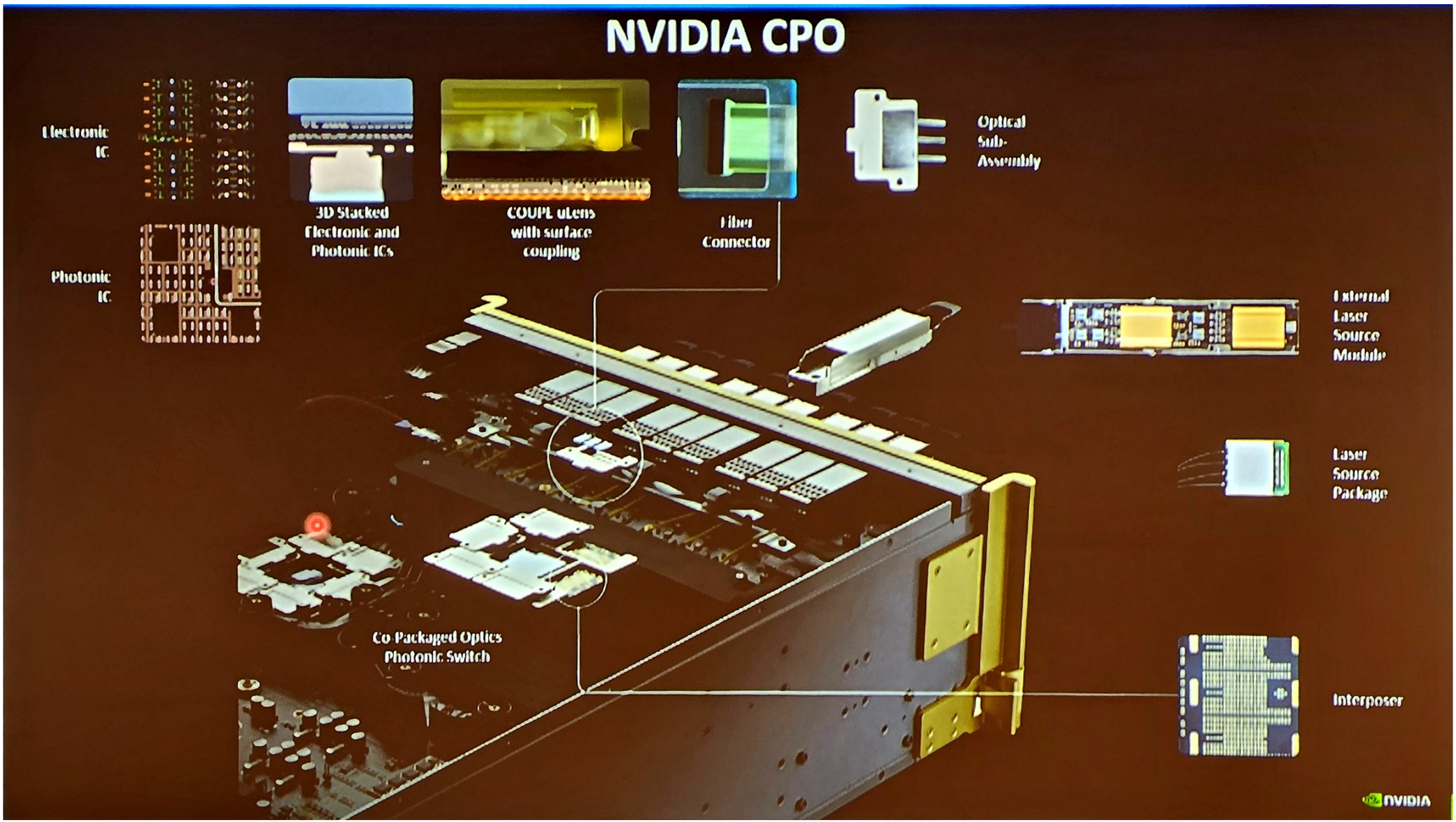
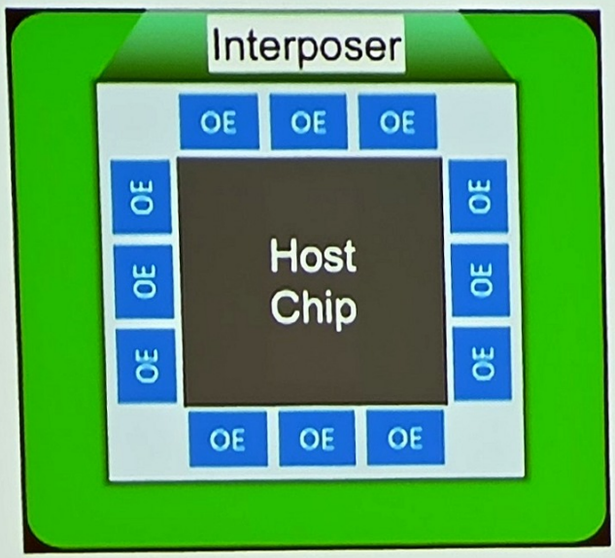
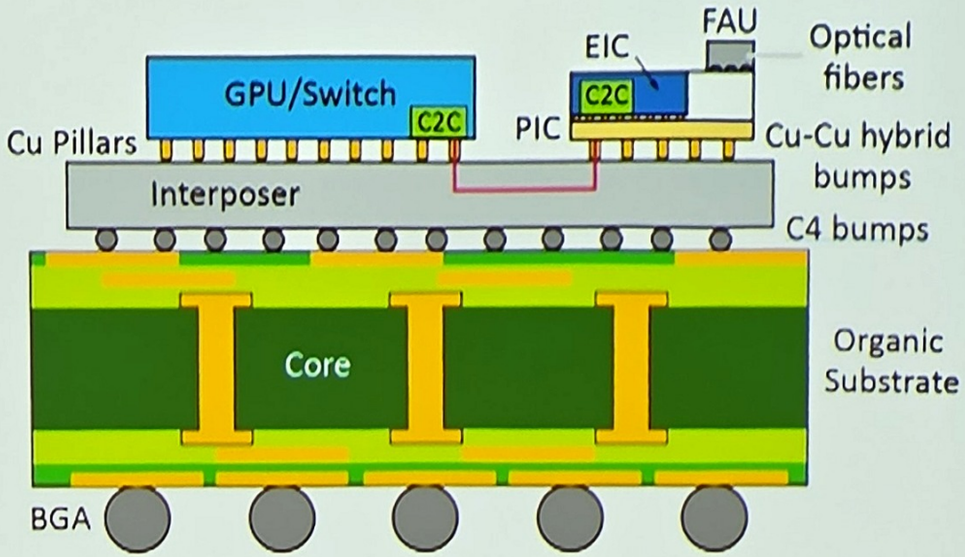


Co-Integrating Optics on Interposer (CIO)

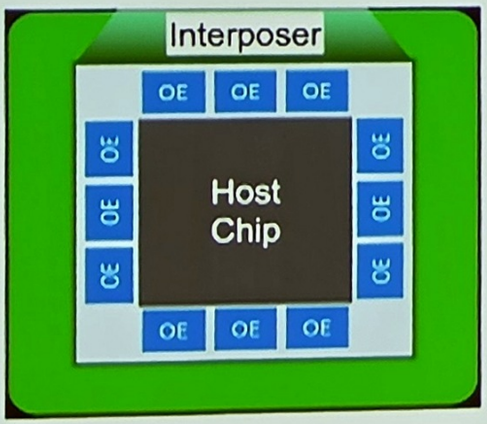




FAU: Fiber array unit. C2C: Chip-to-chip OE: Optical Engine (EIC/PIC/FAU)

Si Interposer offers dense routing → Extend the reach of C2C USR
Benefits of CIO: (1) No beach front expansion. Lowest electrical serdes power. (2) Improved latency (3) simplifies assembly and packaging.

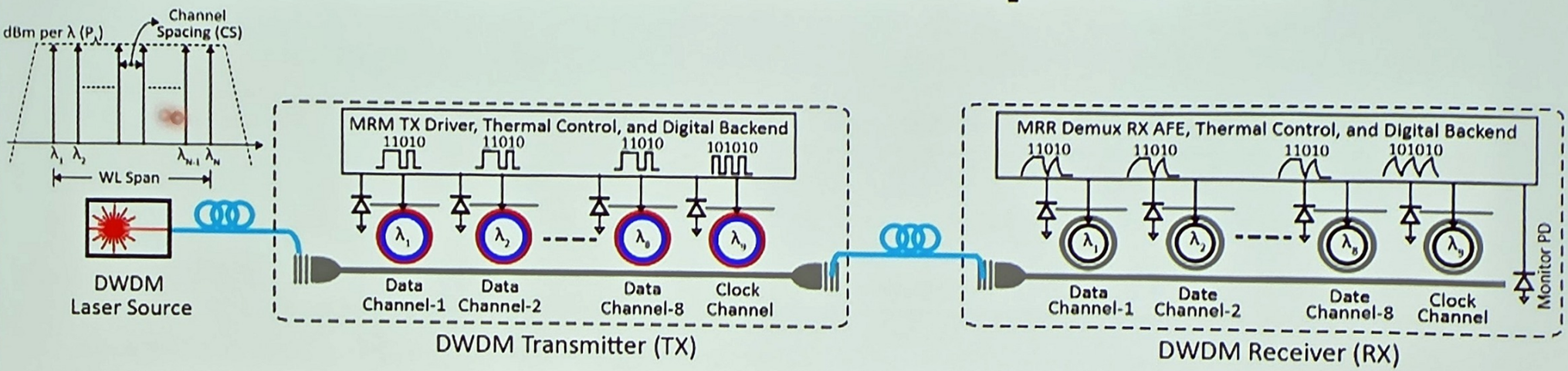
CIO Implementation Challenges



1. High areal bandwidth density needed ~2.5 mm2/Tb/s (full link) → compact photonic components.
2. Small beachfront expansion requires high edge bandwidth density of optical I/O.
3. Compact design increases power density.
4. Thermal envelope requires low-power design.
5. Integration using interposer can be expensive.
6. Fiber array units (FAU) and lens assembly further increase the cost.

1. Area efficient design
2. Low-power consumption
3. Reduced cost

Clock-forwarded DWDM Optical I/O



- DWDM laser source outputs optical power on multiple wavelength.
- Uses MRR-based modulators (MRM) at TX and MRR filters at RX.
- Clock-forwarding allows wideband jitter tracking between data and clock lanes.
- Simplifies circuits and reduces design complexity → minimizes area and power
- Uncorrelated noise between the clock and data lane is **NOT** tracked.

Challenge: Forward half-rate clock while minimizing uncorrelated jitter

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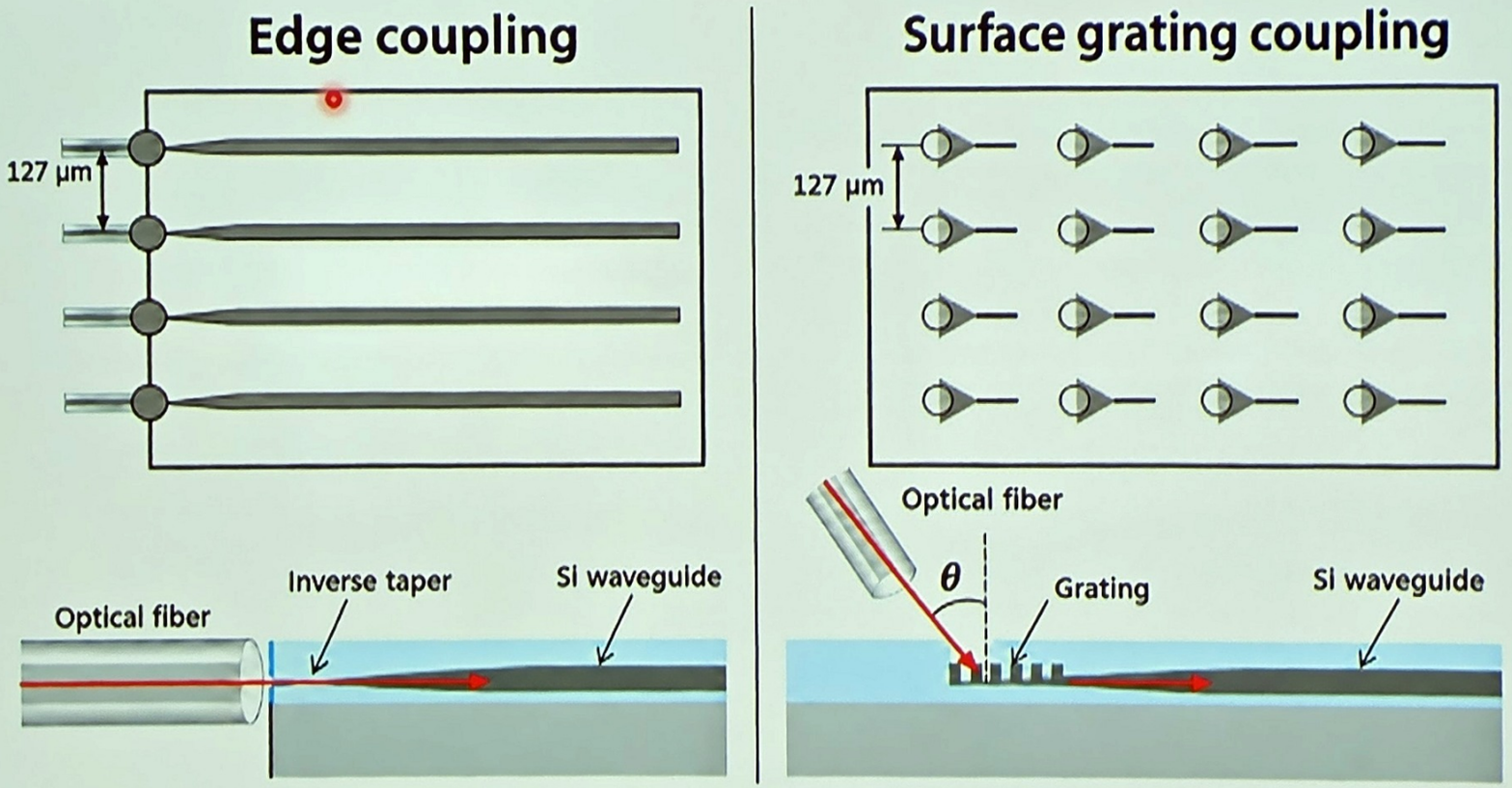
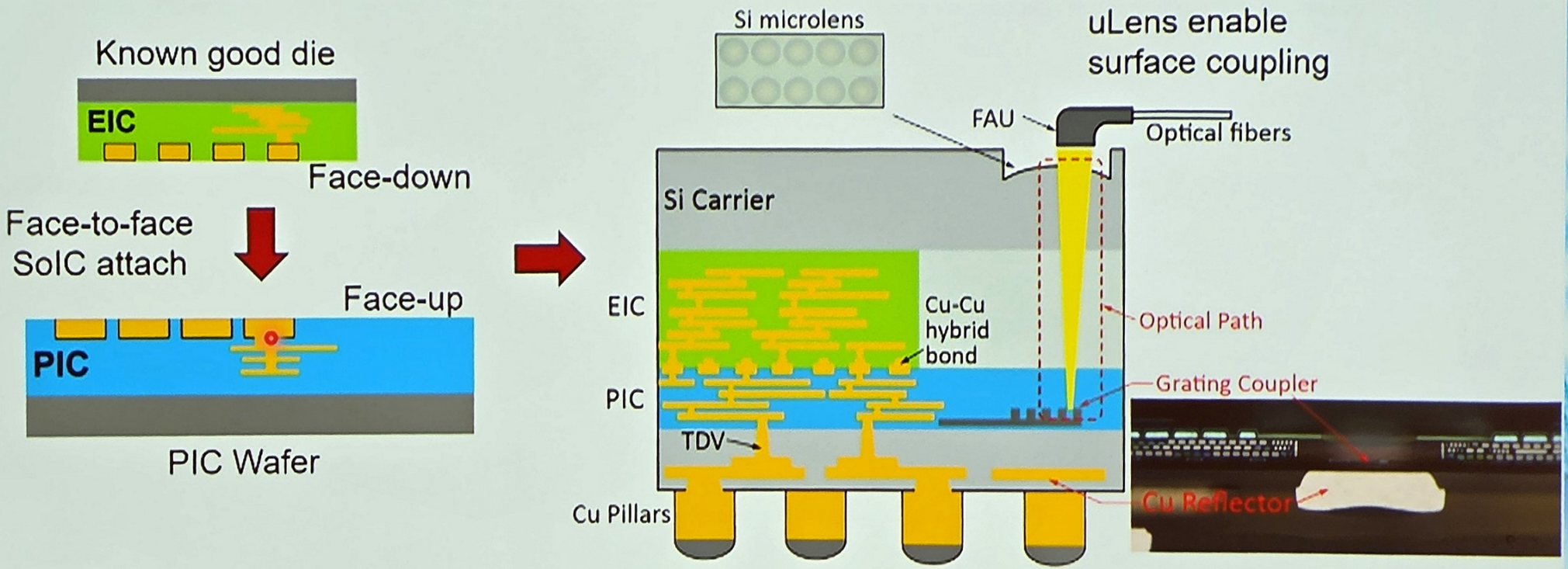
Optical I/O Coupling



- Edge coupling accesses the chip perimeter.
- Surface grating coupling enables a 2D array of fiber connections.

3-D Stacked EIC/PIC SiPh Platform

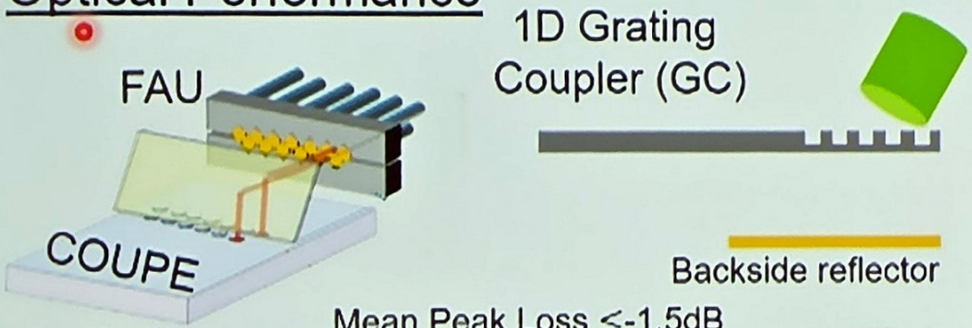
- Based on TSMC COUPE → leverages Cu-Cu hybrid (SoIC) bonding
- EIC is 7nm FinFET CMOS and PIC is 65nm SOI SiPh

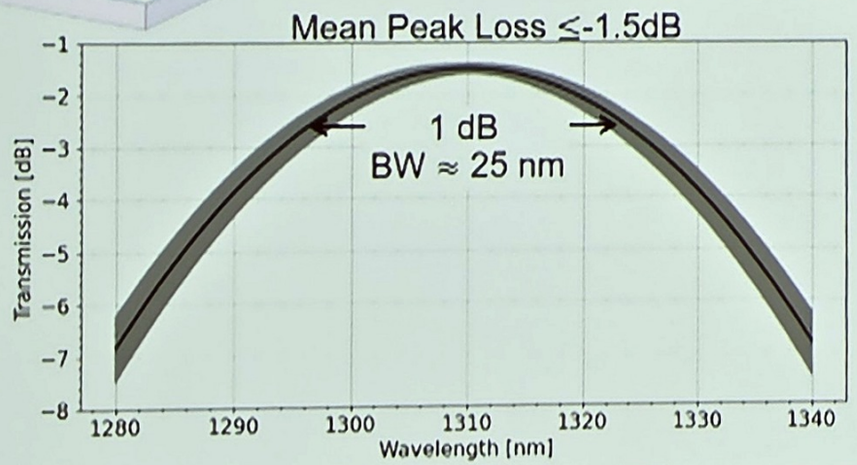


[M.F. Chen ECTC 2025]

3-D Stacked EIC/PIC SiPh Platform

Optical Performance



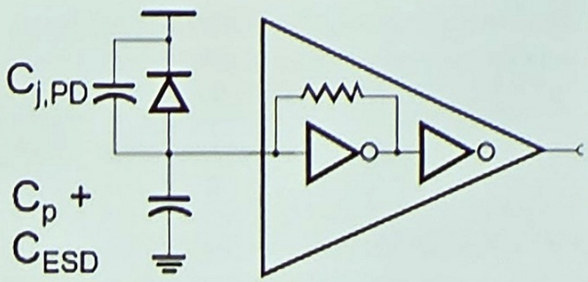


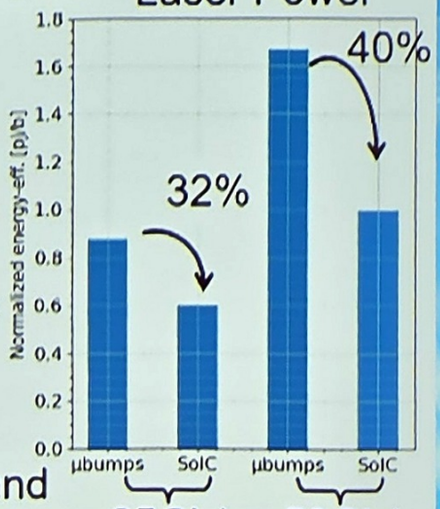
- 1D GC loss <-1.5 dB with ≤ ±1.7nm spread
- Metal reflector improves coupling by ~0.5 dB

GC surface coupling enables high-BW density

Electrical Performance

Ex-1: TIA sensitivity



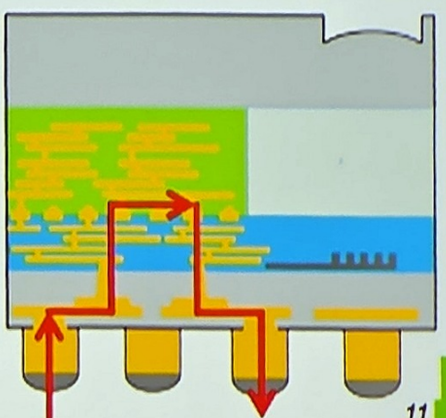


[C.H. Lu ECTC 2024]

- COUPE reduces C_p + C_{ESD} and $C_{j,PD}$ → Improves RX sensitivity

Ex-2: TDV/SoIC RF BW

- Low resistance of TDV and SoIC bonds enables ~100 GHz of RF BW.



Lower parasitic reduces link power consumption

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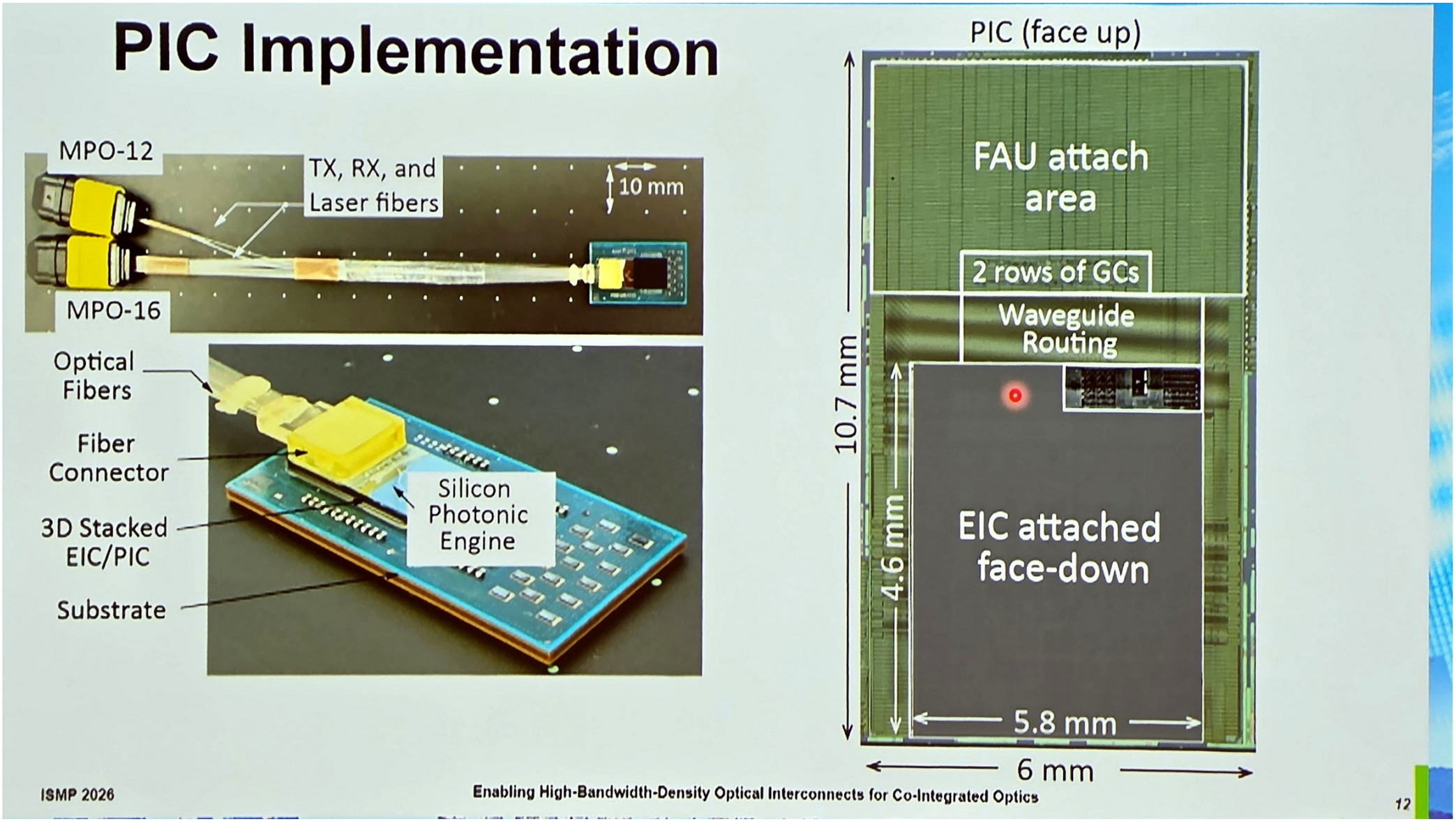
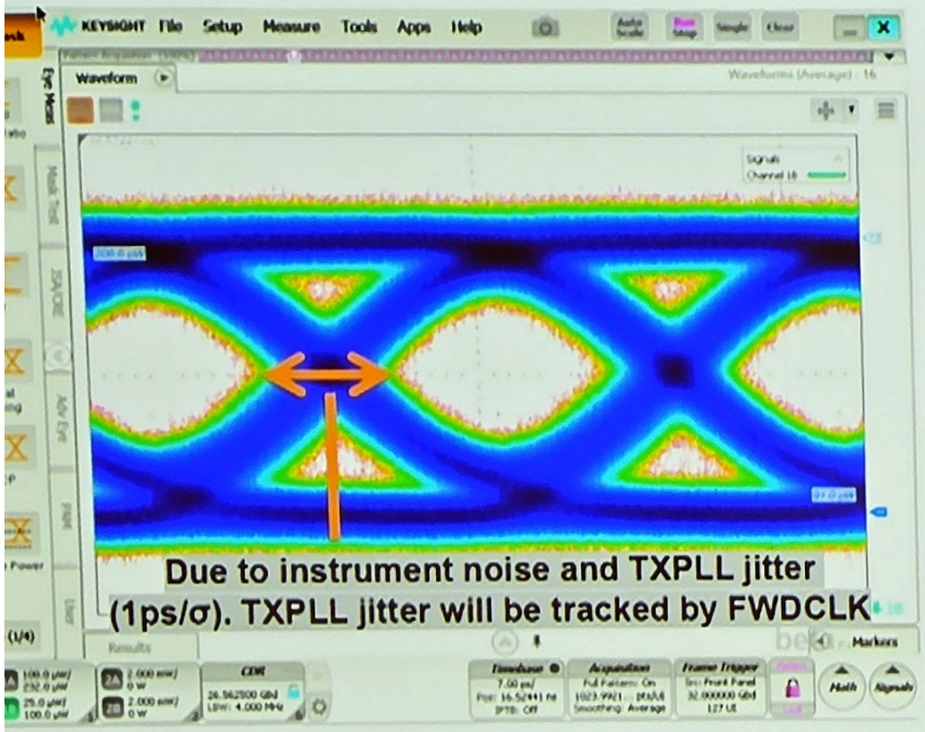
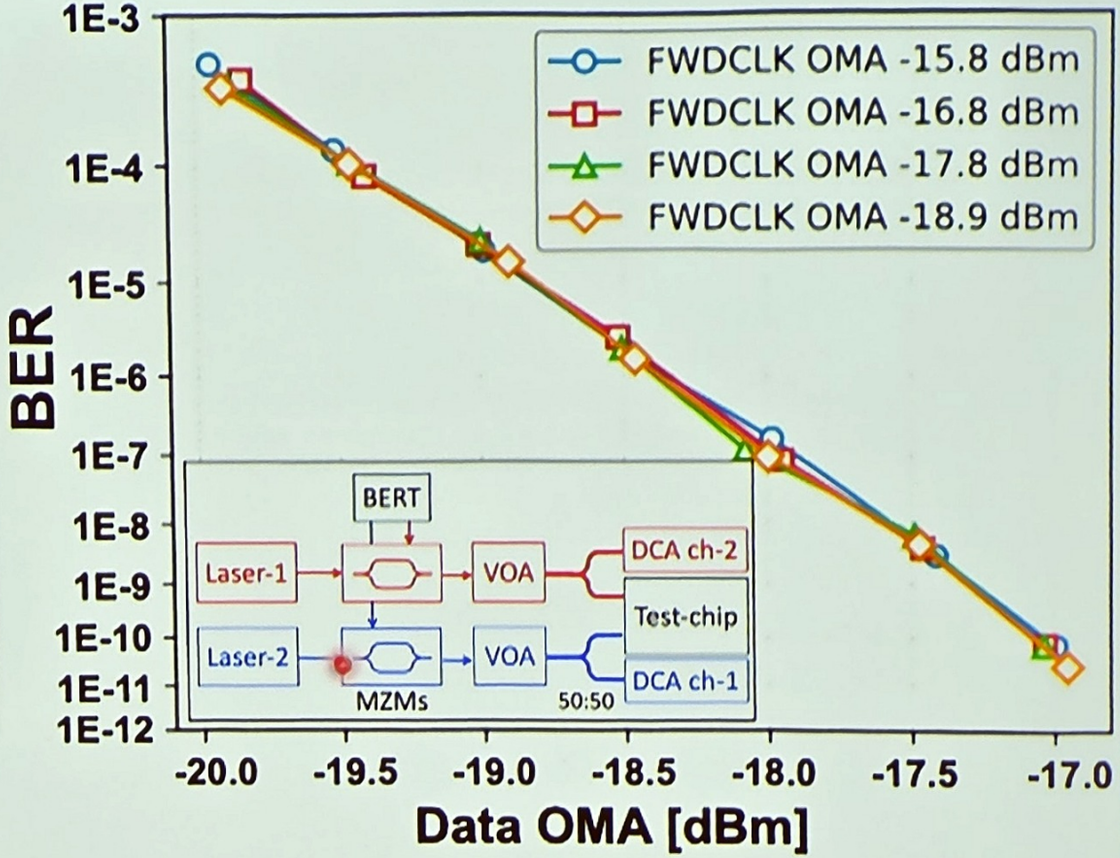


TX and RX Characterization

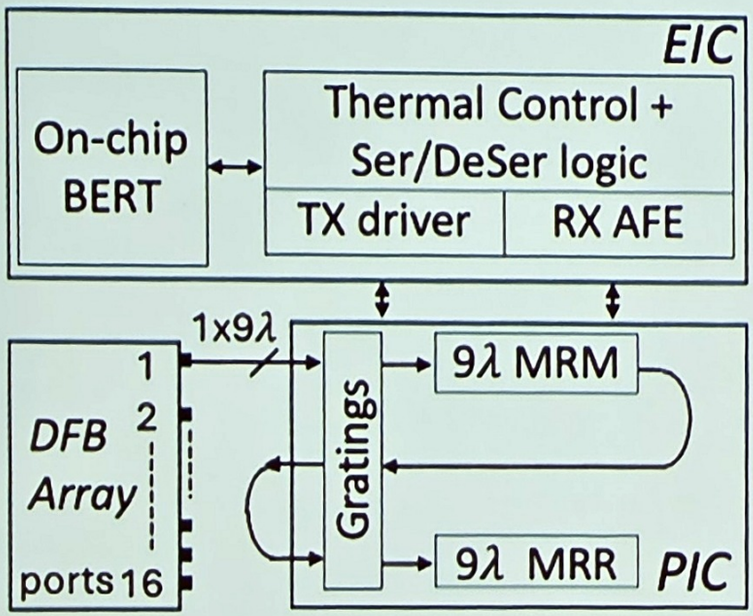


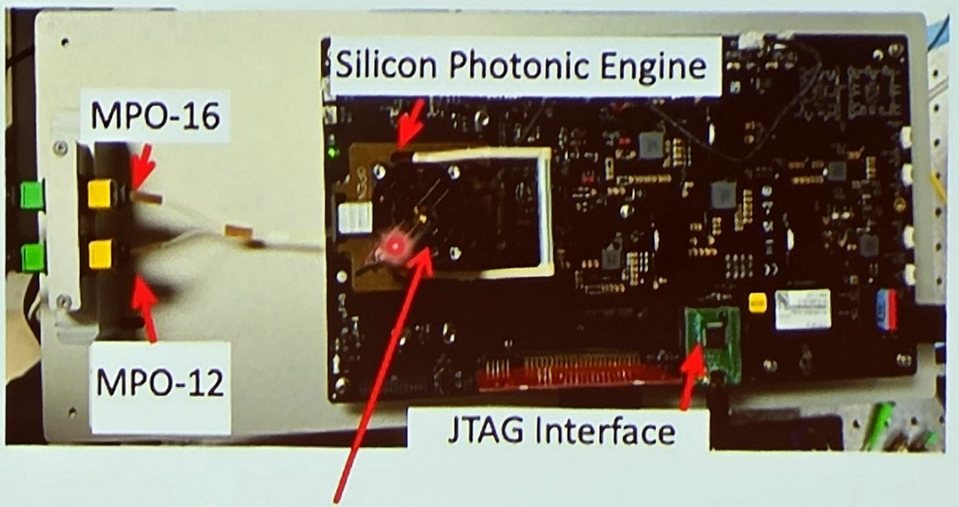


PRBS7 pattern @ 32 Gb/s
TX eye captured with averaging
MRM biased to achieve ER = 3.6 dB

- RX sensitivity <-16.6 dBm at BER of $<10^{-12}$

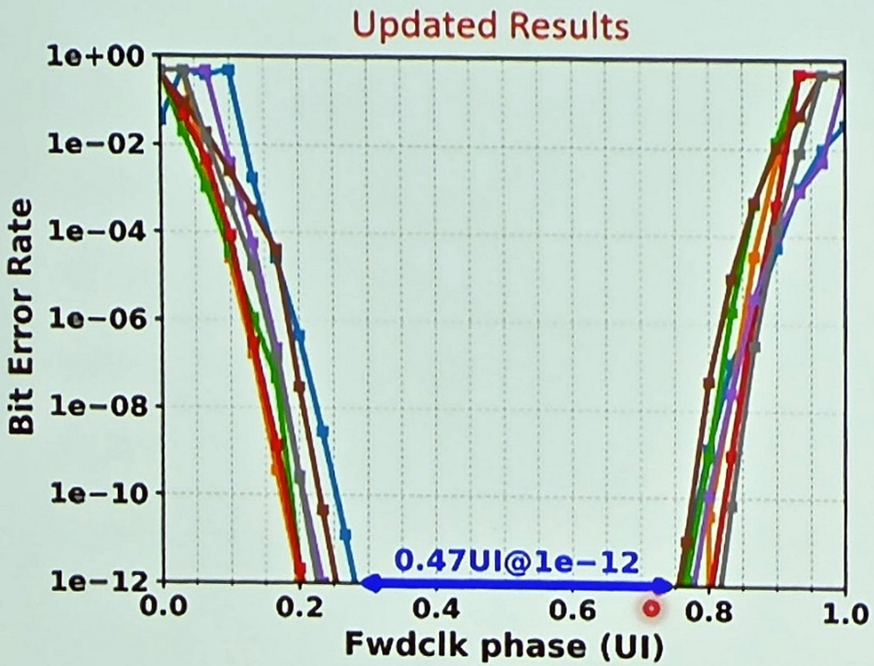
Loopback @ 32 Gb/s: Test-bench





- MPO-12 carries the laser input and MPO-16 carries the DWDM TX and RX.
- Loop-back applied on the same chip with an external fiber connection.
- Two types of laser sources are used: (1) Lab bring-up grade (2) Commercial production-grade

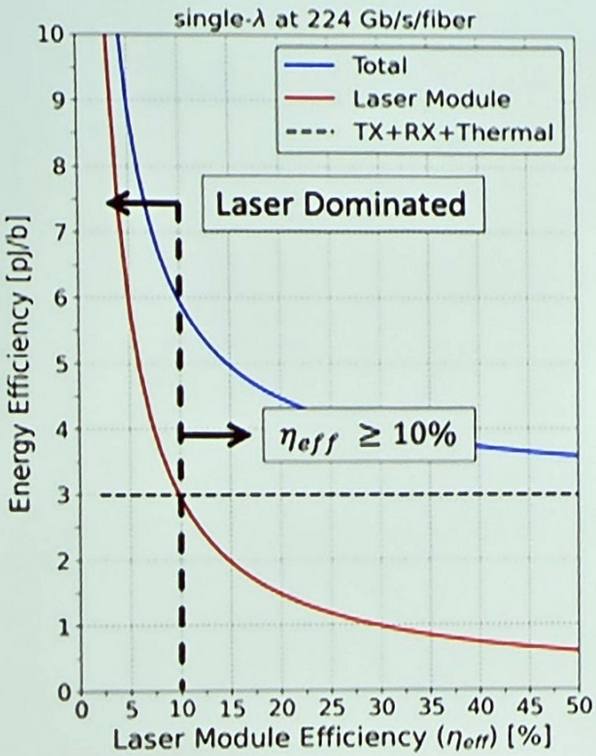
Loopback @ 32 Gb/s: Bathtub and BER

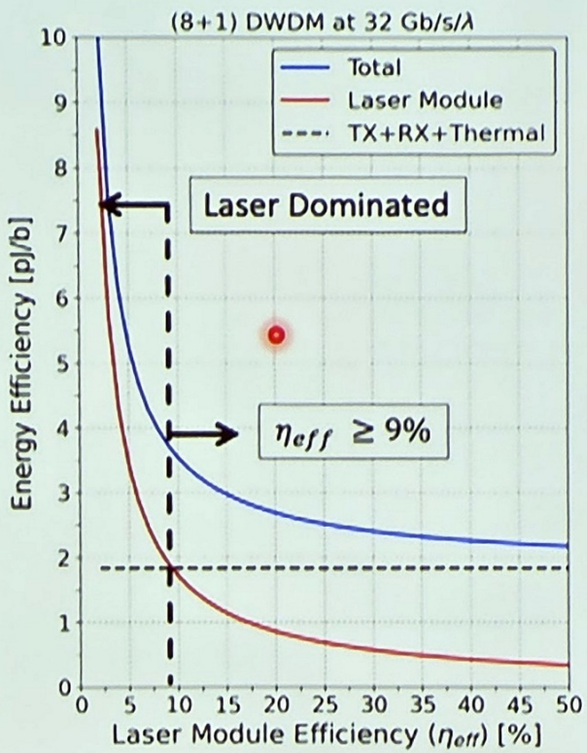


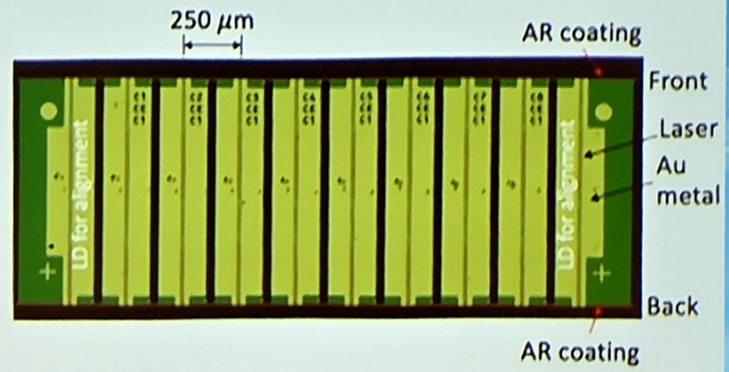
Summary of 8 Data-Lane BER			
Lane	Total Bits	Err Count	BER
TX0 → RX4	2.75e+15	0	<3.6e-16
TX1 → RX5	2.75e+15	0	<3.6e-16
TX3 → RX7	2.75e+15	0	<3.6e-16
TX4 → RX8	2.75e+15	0	<3.6e-16
TX5 → RX0	2.75e+15	0	<3.6e-16
TX6 → RX1	2.75e+15	16	5.82e-15
TX7 → RX2	2.75e+15	0	<3.6e-16
TX8 → RX3	2.75e+15	0	<3.6e-16
Total	**2.2e+16**	**1.6e+01**	**7.27e-16**

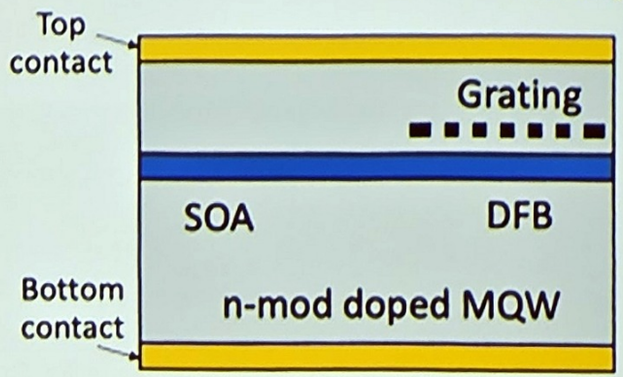
- All lanes **ON** and **TT** locked. PRBS31 test-pattern used
- 0.47 UI aggregate eye opening @ BER $<10^{-12}$
- Long-term BER measurement: Stable link operation over 24+ hours of interval

Laser Module Efficiency









For both cases: $\eta > 10\%$

1. N-mod-doped MQW + AR/AR coating
2. SOA boosts output and acts as spot-size converter

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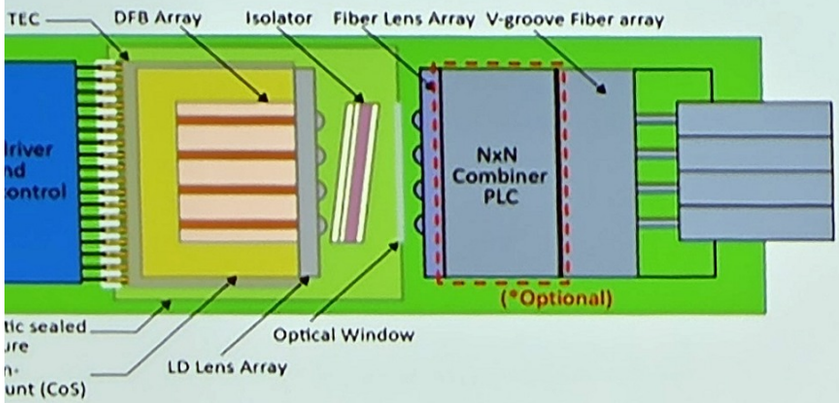
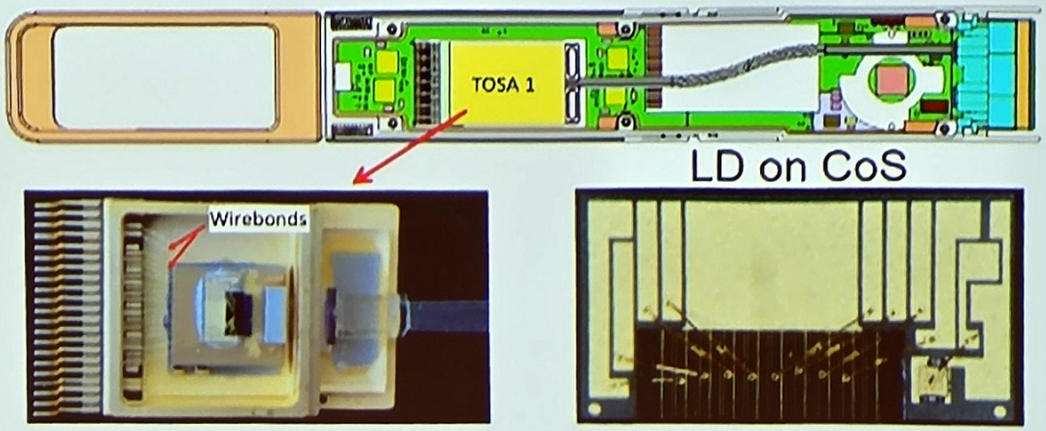
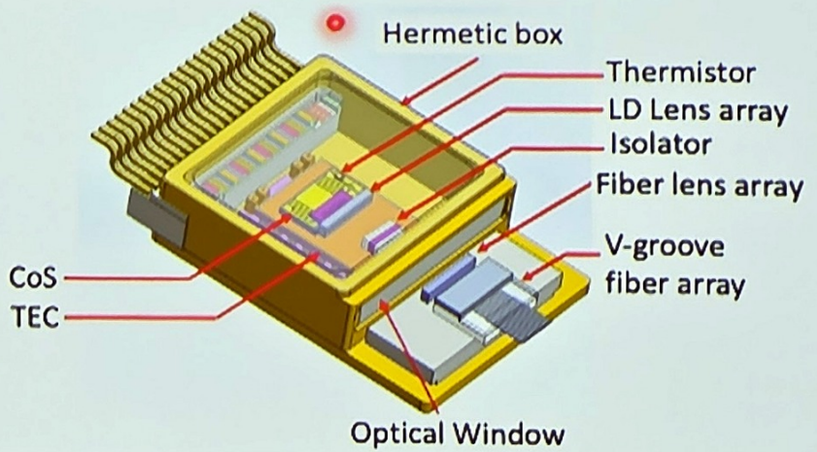
DFB-Array LD in ELSFP module





Key packaging moves

- Single-step epoxy cure
- Array-level auto-alignment
- Lasers ON during alignment
- Hermetic Au box + TEC

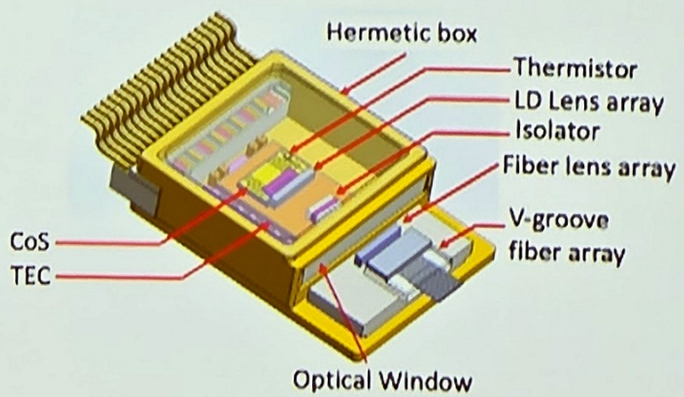


Measured DWDM laser performance

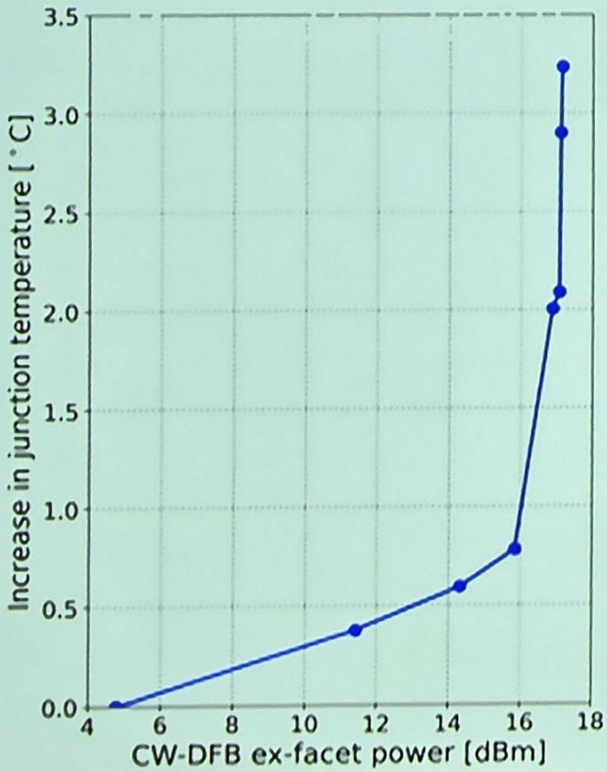
< ±0.8 dB	< ±16 GHz	7.2%	< −134 dB/Hz
ΔP across channels	CS error	module efficiency	RIN, all channels

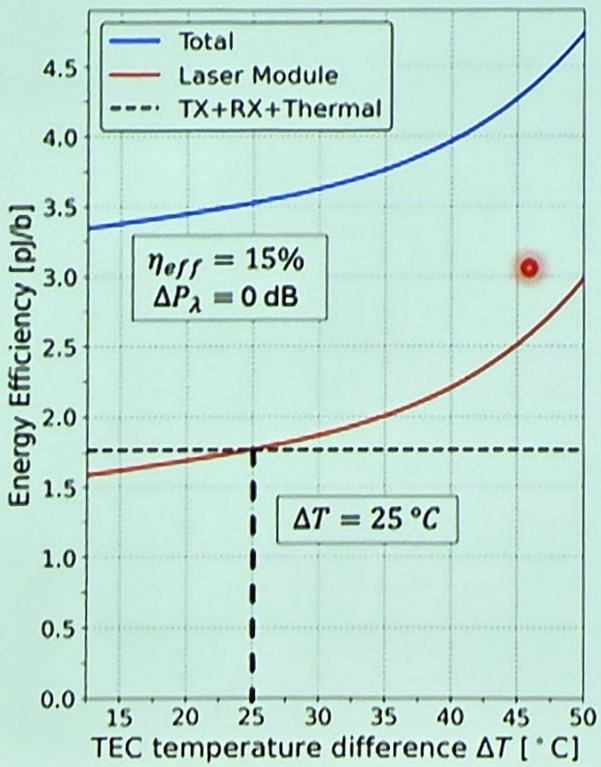
Measurement conditions:

- 8 channels at 200 GHz nominal spacing
- Power variation measured in fiber after 8×8 mux
- 7.2% module efficiency excludes TEC power



Model: Impact of TEC Power

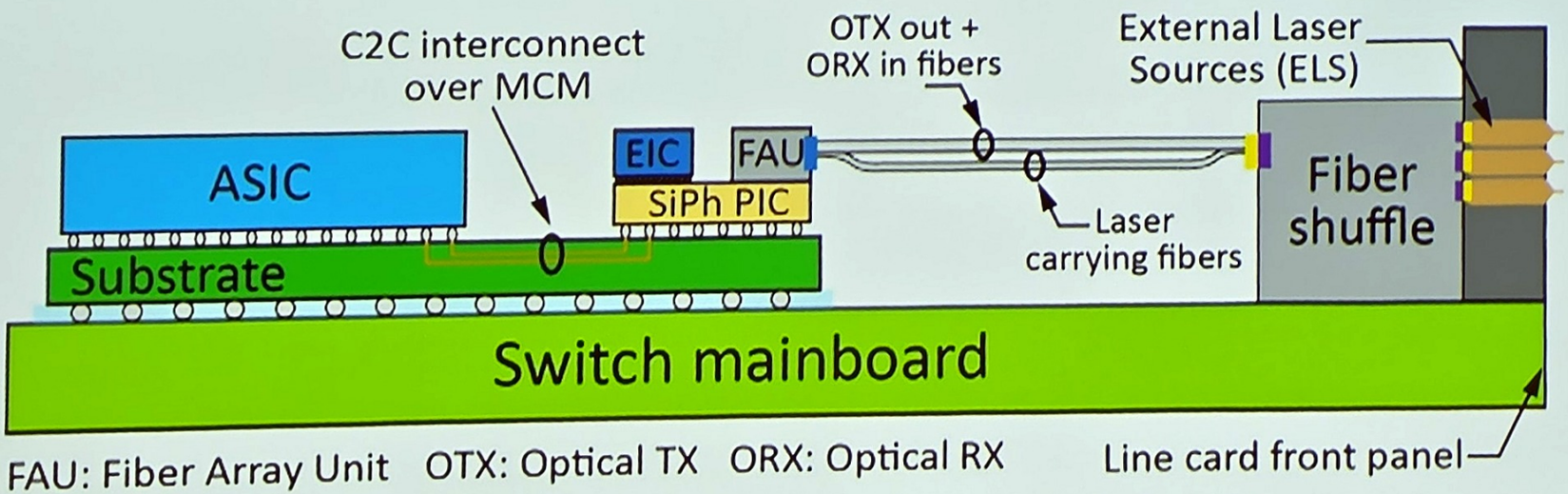




- Higher laser power raises junction temperature
- Higher temperature difference increases TEC power
- For this model, TEC dominates link energy at ΔT > 25 °C

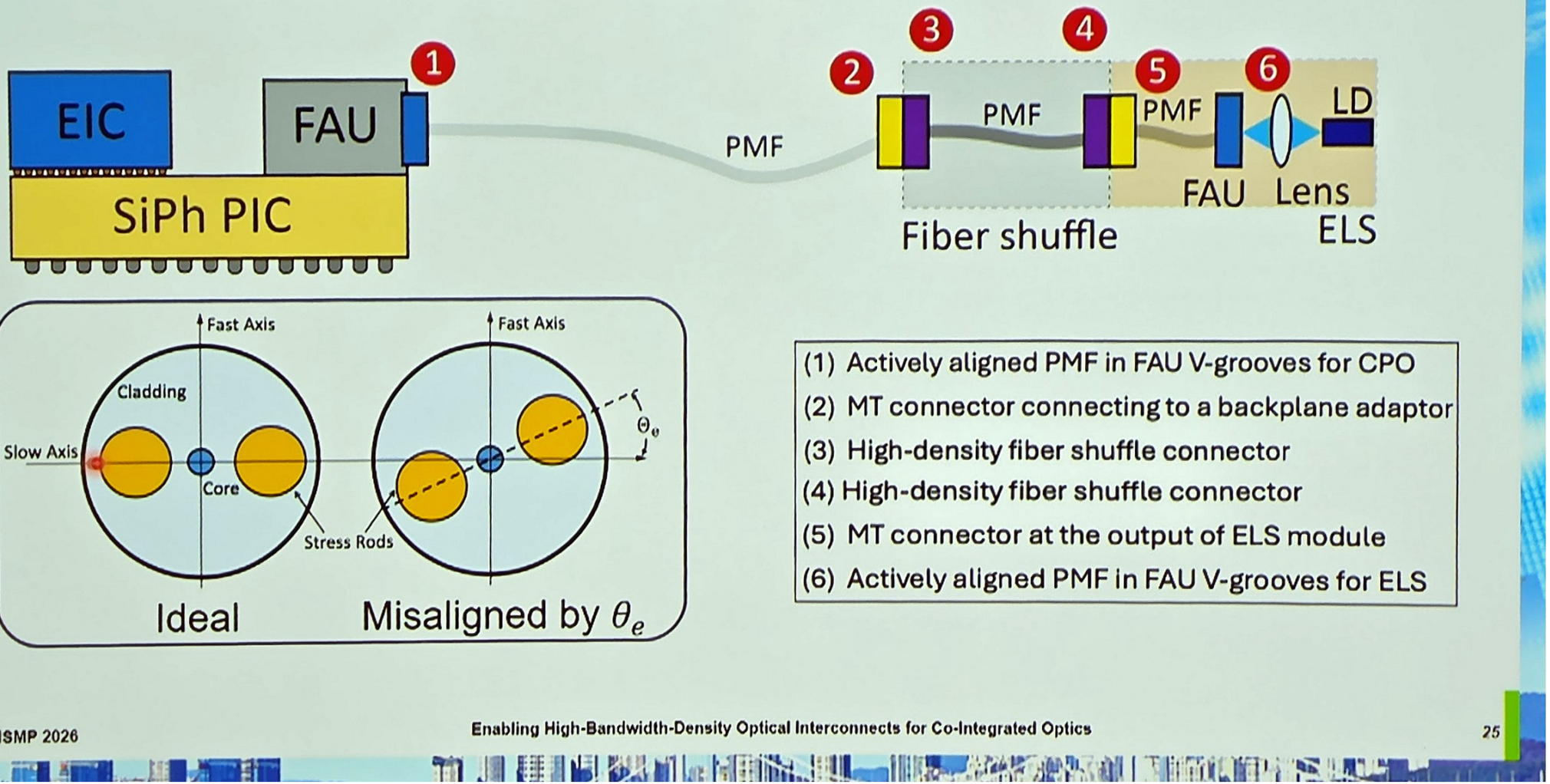
1. Efficient DWDM lasers
2. Avoid TEC by moving cooling burden to liquid cool systems

Simplified View of ELS-CPO Optical path



- Two main types of fibers: (1) Laser carrying PMFs (2) TX/RX data carrying SMFs
- Fibers are well secured. No dangling fiber. Shuffle box ensures correct connection is made between front panel and back plane adapters.
- PMFs to SMFs are typically 1:4 → The total cost is nearly the same.
- Further PMFs require accurate alignment to avoid excess IL and degrading PER

PMF Rotational Misalignment θ_e



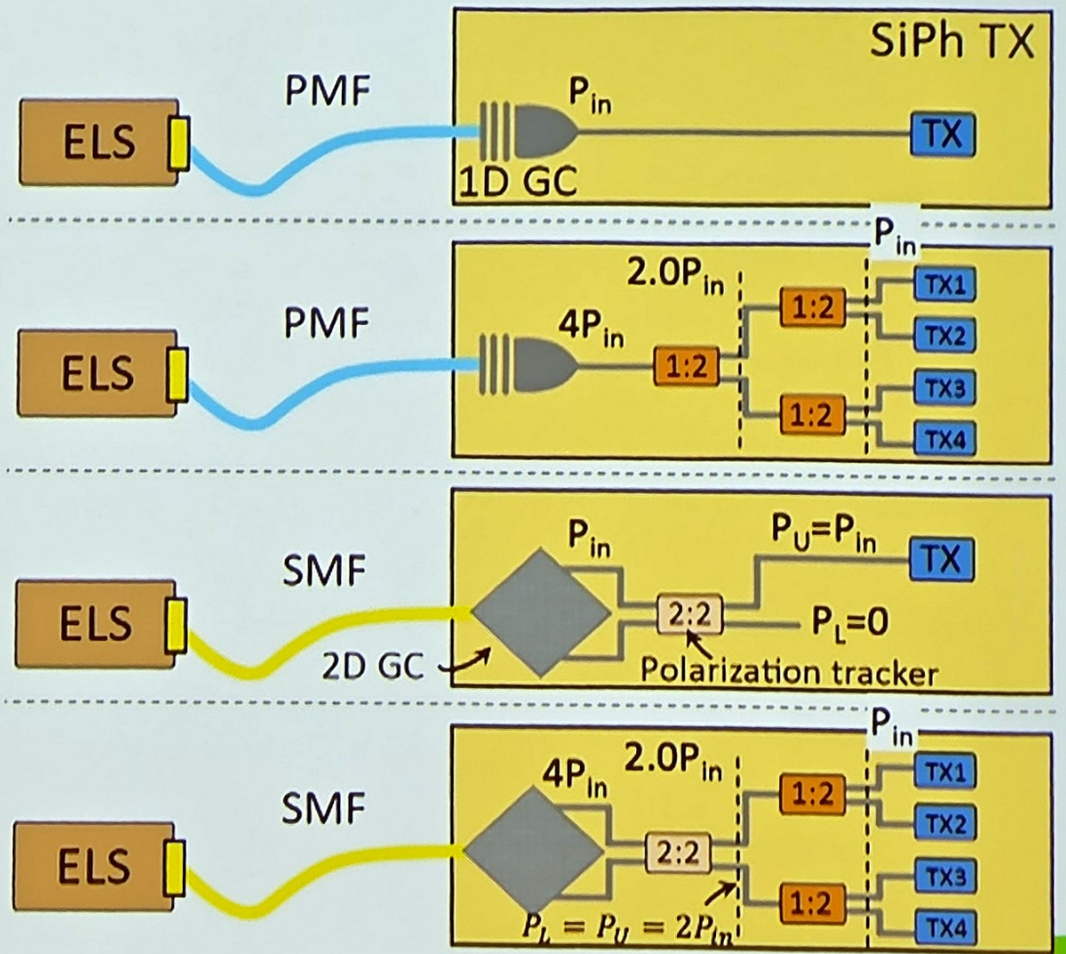
ELS and OTX Connection Scenarios

ase-1: 1 ELS powers 1 OTX.
- Bare minimum → Baseline.
- Expensive → Many ELS needed.

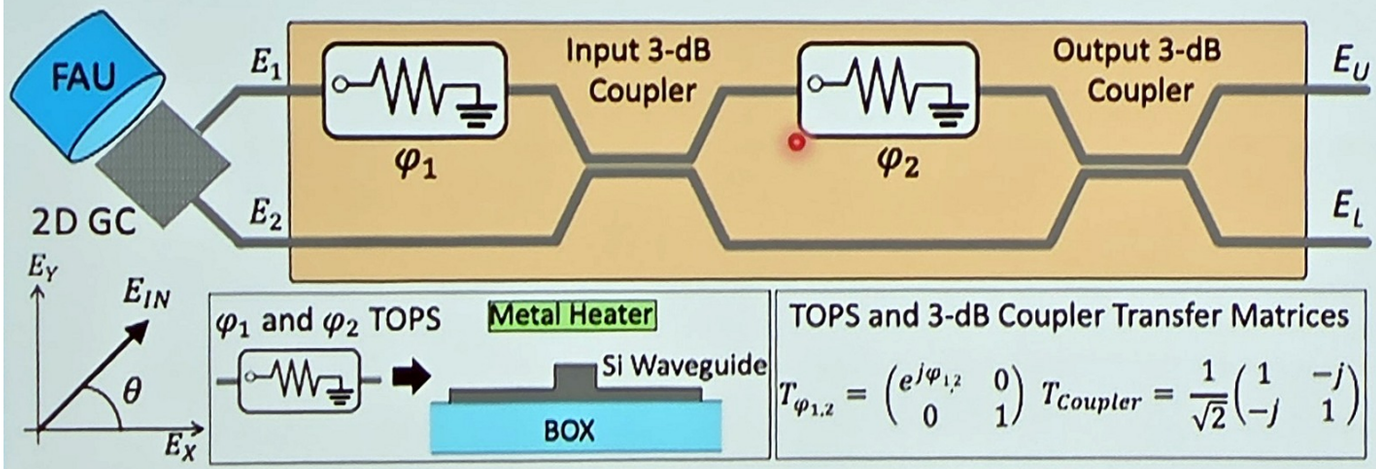
ase-2: 1 ELS powers 4 OTX.
- Commonly used architecture.
- Optimizes ELS cost, efficiency, etc.

ase-3: 1 ELS powers 1 OTX.
- Similar in operation as case-1.
- Expensive and inefficient.

ase-4: 1 ELS powers 4 OTX.
- Preferred architecture.
- Similar in operation as case-2.

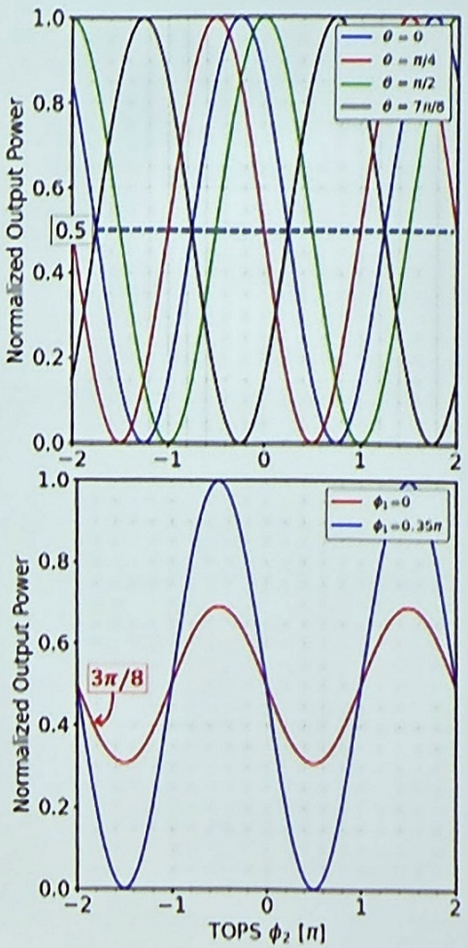


SiPh PIC with Polarization Locking Circuit

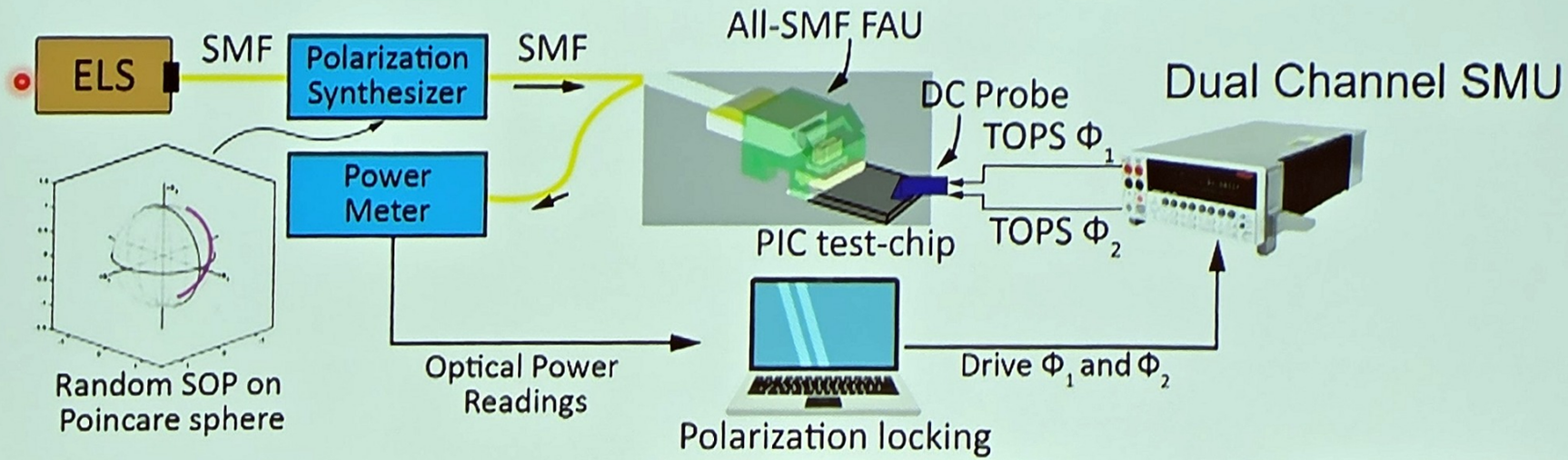


$$\begin{bmatrix} E_U \\ E_L \end{bmatrix} = T_{Coupler} \cdot T_{\phi_2} \cdot T_{Coupler} \cdot T_{\phi_1} \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}$$

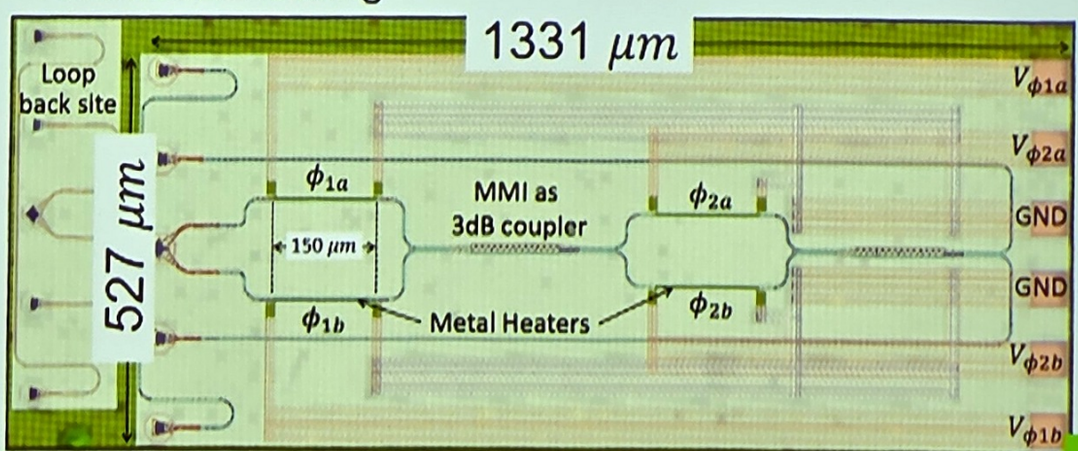
- Thermo-Optic Phase Shifters (TOPS) add phase-shift by thermally changing η_{eff}
- ϕ_1 and ϕ_2 convert random input SOP to TE



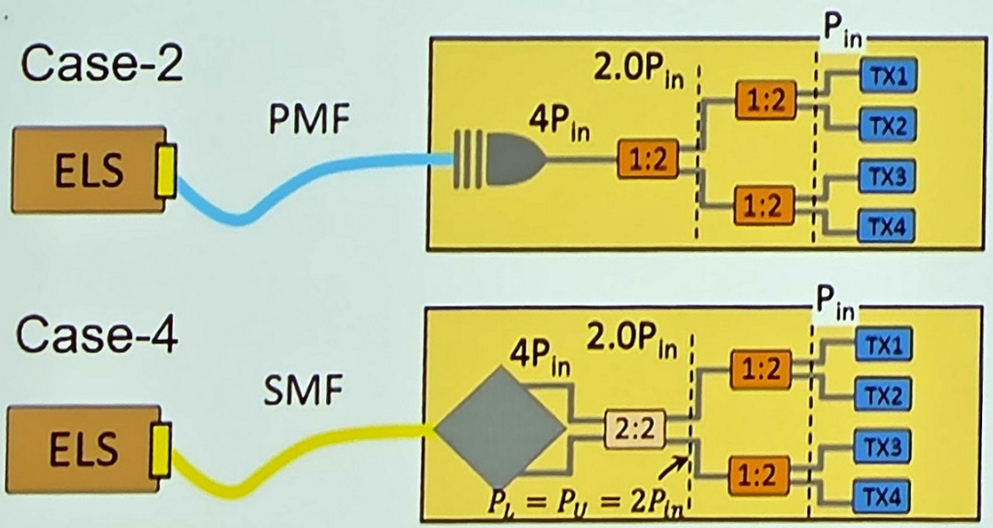
Test-bench for Pol. Locking SiPh test-chip

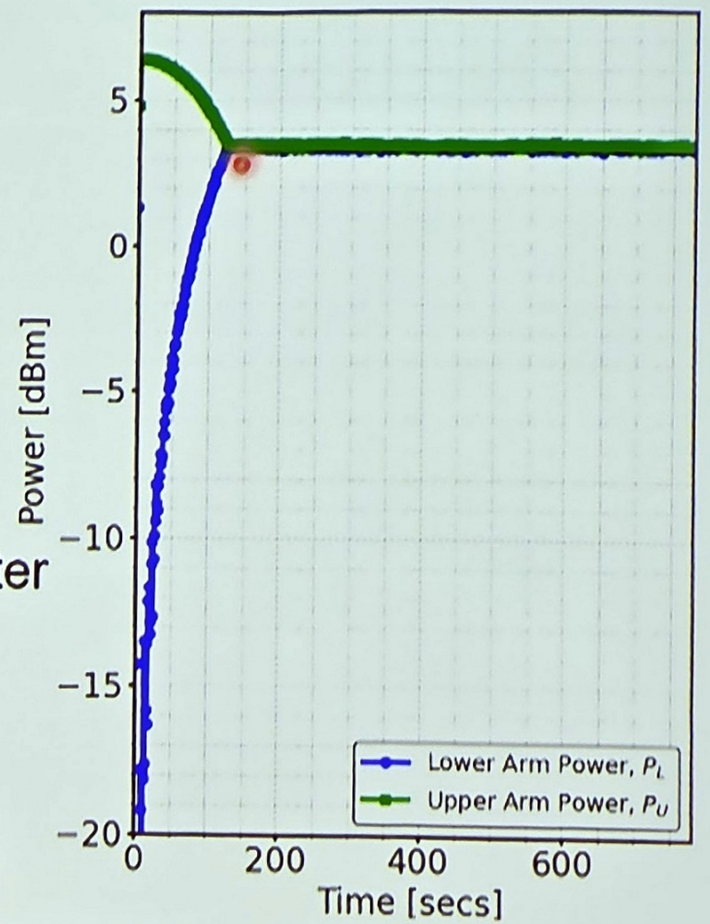


- Polarization synthesizer injects random SOP at the PIC input
- Pol locking algorithm → GD search
- Dual channel SMU drives TOPS
- PIC designed in TSMC SiPh Process



Measured Optical Power Under Closed-Loop

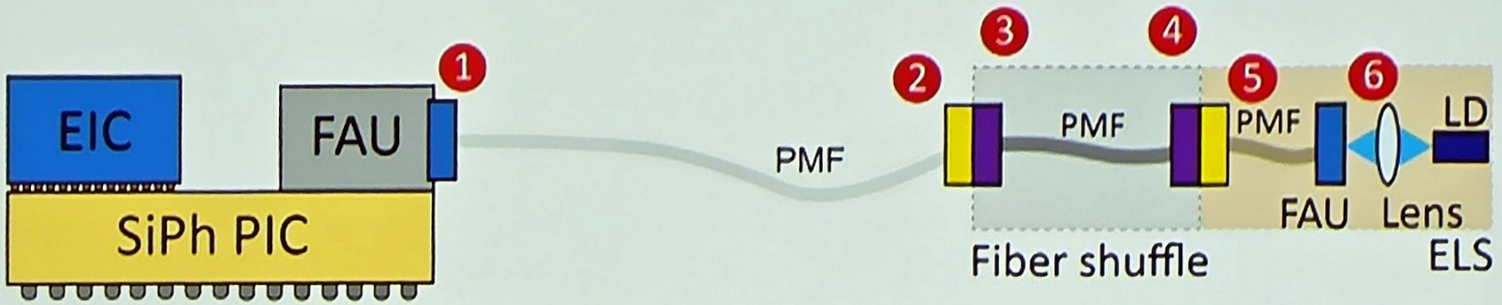




- Pol. locking circuit absorbed as one of the power splitter
- Equalizes the power in two arms → $P_U = P_L = 0.5*P_{in}$
- Amortizes 2D GC IL across 4 OTX
- Well-matched layout → Needs only 1 TOPS ϕ_2

Lowest IL penalty and pJ/b overhead

Cost Breakdown

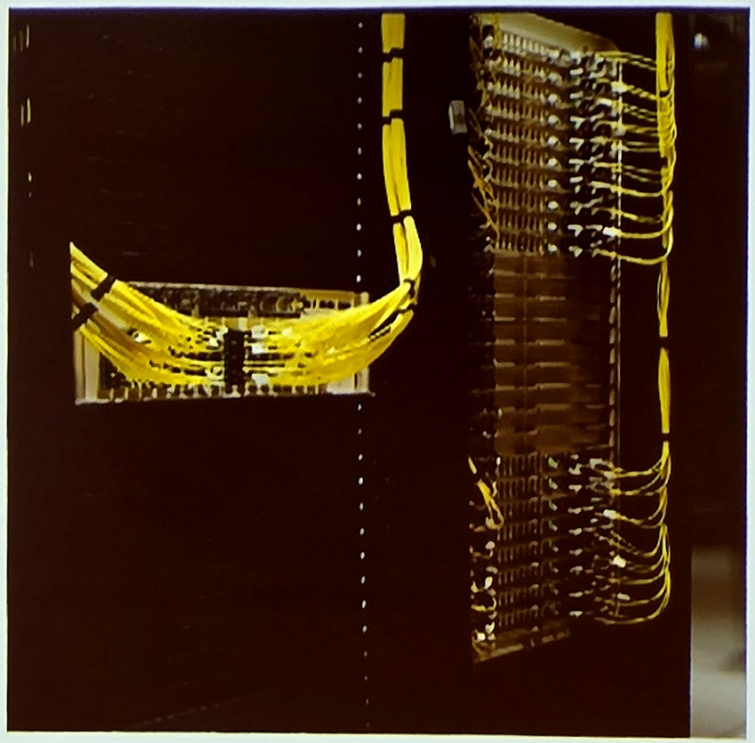


No	Component	Location*	Cost of PMF/Total SMF Cost
1	Fiber Ribbon (2meter)	(1), (2)	25.6%
2	Assembly + Termination‡ (x2 MT)	(2), (5)	33.3%
3	Fiber Shuffle‡ (ELS MT-MT)	(3), (4)	161.5%
4	FAU with V-groove (x2)	(1), (6)	116.15%
		Total	271.3% (~2.7x)

*Along the optical path of Figure 4. ‡Function of rotational alignment accuracy

Future Scaling

- Scaling data rate 32 Gb/s → 64 Gb/s (sub-2nm process)
- Polarization & Optical BiDi provides another 2x scaling
- Multi-wavelengths
  - DWDM:16λ, 32λ and beyond
  - CWDM: 2x400Gbps, 4x400Gbps
- Coherent lite
- Need to strike the right balance between
  - BW per fiber
  - Radix
  - Total throughput



NVL72 with Quantum-X InfiniBand Photonics Scale-Out

Key Takeaways

1. **Co-integrated optics** enables high-bandwidth-density connections while reducing pressure on package shoreline.
2. **DWDM microring optical I/O** demonstrated 256 Gb/s per fiber at 2.6 pJ/bit using a 3D-stacked electronic and photonic platform.
3. **Multiwavelength laser packaging** enables an eight-channel source with 200 GHz nominal spacing; laser efficiency and cooling remain important to the link energy budget.
4. **Fiber coupling and attachment** are central packaging choices. Use of surface coupling + All-SMF FAUs enable high-BW density at low cost.

Future scaling (BiDi, Pol Mux, etc ...) will depend on advances in laser efficiency, microring thermal control, fiber assembly, and package integration.